

SERVICE NOW VIRTUAL AGENT FOR AUTOMATED IT AND BUSINESS SUPPORT

A Dissertation

*Submitted in partial fulfillment of the requirements for the award of the
degree of*

BACHELOR OF TECHNOLOGY

IN

Department of CSE - ARTIFICIAL INTELLIGENCE

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DEPARTMENT OF CSE – ARTIFICIAL INTELLIGENCE

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Vinjanampadu (V), Vatticherukuru (M), Guntur (Dt), A.P-522017.

APRIL – 2025.

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CERTIFICATE

This is to certify that this project report entitled **“SERVICE NOW VIRTUALAGENT FOR AUTOMATED IT AND BUSINESS SUPPORT”** submitted by **G. Sanavya (22JR5A4305), J. Naga Sri Jyothi (21JR1A4367), M. Pushpa Sai Sri Guneswari Devi (22JR5A4306)** to Jawaharlal Nehru University Kakinada, through KKR & KSR Institute of Technology and Sciences for the award of the Degree of Bachelor of Technology in department of CSE - Artificial Intelligence is a bonafide record of project work carried out by them under my supervision during the year 2024 - 2025.

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DECLARATION

We hereby declare that the project **“SERVICE NOW VIRTUAL AGENT FOR AUTOMATED IT AND BUSINESS SUPPORT”** has been carried out by us and this work has been submitted to KKR & KSR Institute of Technology and Sciences (A), Vinjanampadu, affiliated to Jawaharlal Nehru Technological University, Kakinada in partial fulfillment of the requirements for the award of degree of Bachelor of Technology in department of CSE - Artificial Intelligence.

We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

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- To ensure the all-around development of students through a judicious blend of curricular, co-curricular, and extra-curricular activities.
- To support the cross-cultural exchange of knowledge between industry and academy.
- To provide higher/continued education and research opportunities to the employees of the institution.



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- Train students to be technically competent through innovation and leadership.
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Program Specific Outcomes (PSOs)

PSO1: Application Development

Apply the concepts in core area of Artificial Intelligence, Data Structure, Database System, Operating System, Networking and Intelligence System to solve futuristic problems.

PSO2: Computing Paradigms

Develop automated solutions for real world problems through laboratory experiments, projects and internship

Program Educational Objectives (PEOs)

PEO:1	Graduates of Computer Science and Engineering – Artificial Intelligence shall apply appropriate theory, practices, and tools to provide solution for multidisciplinary challenges.
PEO:2	Graduates of Computer Science and Engineering - Artificial Intelligence shall have an ability to function effectively in the workplace for professional growth.
PEO:3	Graduates of Computer Science and Engineering shall have exposure to adapt, contribute and innovate new technologies in the key domains of Artificial Intelligence during higher studies or product development.



PROGRAM OUTCOMES (POS)

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with and Understanding of limitations.
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.



- 10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- 11. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- 12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Outcomes (COS)

S. No.	B. Tech Project CO
CO 1	Interact with customers and identify real world problem statement / identify problems in engineering and technology in selected field of interest.
CO 2	Synthesize and apply prior knowledge of mathematics, computer science and engineering to design and implement solutions to open-ended problems.
CO 3	Design and Develop the software with Software Engineering practices and standards.
CO 4	Use different tools for communication, design, implementation, testing and report writing.
CO 5	Analyzing professional issues, including ethical, legal and security issues, related to software project.
CO 6	Develop better interpersonal communication skills, team work and leadership qualities with writing and oral presentation skills.

Course Outcomes - Program Outcomes mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 10	PO 12	PSO 1	PSO 2
CO421.1	1	2							1	3				2
CO421.2	2	2	1	3								1		3
CO421.3			2		2	3	1	1					3	
CO421.4				1	3	2	2			2	1		2	
CO421.5						1	2	3						
CO421.6									3	1	2	2		

3: High 2: Medium 1: Low

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LIST OF ABBREVIATIONS

ITOM	Information Technology Operation Management
CSM	Customer Service Management
ITSM	Information Technology Service Management

ABSTRACT

This project focuses on implementing the ServiceNow Virtual Agent to automate IT and business service requests. The Virtual Agent leverages AI and Natural Language Understanding (NLU) to handle user queries efficiently. It reduces manual workload by automating routine tasks like incident management and service requests. Custom conversation flows are designed to improve user interaction and response times. Integration with ServiceNow's backend enhances workflow automation. The project also analyzes chatbot performance for continuous improvement. The solution ensures faster resolution of issues with minimal human intervention. It enhances user satisfaction by providing instant and accurate responses. The implementation is scalable for enterprise use. Overall, it streamlines IT and business operations efficiently.

The ServiceNow Virtual Agent is an AI-powered chatbot designed to automate IT and business support services. It leverages Natural Language Understanding (NLU) to interact with users, handle service requests, and resolve issues without human intervention. By integrating with the ServiceNow platform, the Virtual Agent enhances workflow automation, reducing the workload of IT and support teams.

This project focuses on implementing the Virtual Agent to improve efficiency in handling routine tasks such as incident management, password resets, service requests, and knowledge-based queries. With customized conversation flows, the chatbot provides an intuitive and responsive user experience. Additionally, real-time analytics and performance monitoring ensure continuous improvement of the system.

Keywords:

ServiceNow Virtual Agent, AI-powered chatbot, Natural Language Understanding (NLU), Workflow automation, Incident management, Password reset automation, Service request handling, Knowledge-based queries, Custom conversation flows, Real-time analytics, Performance monitoring, Continuous improvement, Enterprise scalability, IT support automation, Intelligent virtual assistant, Automated service desk.

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CHAPTER – 1

INTRODUCTION TO SERVICE NOW

1.1 What is ServiceNow?

ServiceNow is a software company that provides a cloud-based, AI-driven platform for automating multiple management workflows in enterprises. The company specializes in IT service management, IT operations management and IT business management.

Founded in 2004, ServiceNow offers an intelligent, cloud-based platform and tools for automating, managing and optimizing many enterprise processes, including the following:

- IT service management (ITSM)

ITSM focuses on delivering and managing IT services to meet business needs effectively. It includes processes like incident management, change management, and service request fulfillment. ServiceNow ITSM streamlines service delivery through automation and AI. This leads to improved efficiency, reduced downtime, and enhanced user satisfaction.

- IT operations management (ITOM)

ITOM ensures the smooth functioning of IT infrastructure and services. It involves monitoring, managing, and optimizing performance across networks, servers, and applications. ServiceNow ITOM uses AIOps to detect issues proactively and automate resolutions. This enhances service reliability and reduces operational costs.

- Customer Service Management (CSM)

CSM is designed to improve customer experiences by resolving issues efficiently. It connects customer service with other departments to ensure end-to-end problem resolution. ServiceNow CSM automates case management and provides self-service options. This results in faster support, better visibility, and improved customer satisfaction.

A cloud-based platform for IT service management (ITSM) and business process automation. Helps organizations streamline operations and improve service delivery.

Provides a wide range of modules designed for different areas like incident management, knowledge management, service catalog, reporting, etc.

1.1.2 ServiceNow Platform capabilities and services



Fig 1.1 : Servicenow capabilities

With ServiceNow, you can choose from four workflows (IT Workflows, Employee Workflows, Customer Workflows, or Creator Workflows).

1.1.3 The ServiceNow Instance

When you are accessing ServiceNow, you are accessing an instance of the Platform. An organization can have several instances (e.g., Development, Quality Assurance (QA), Test, and Production) and each instance is a single implementation of the ServiceNow Platform.

Releases

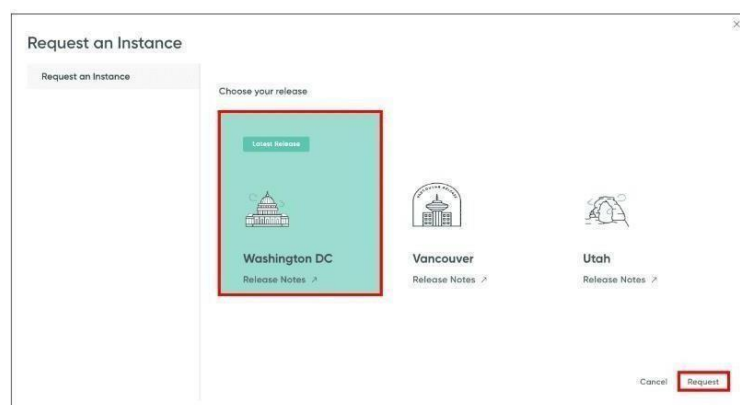


Fig 1.2 Releases

Release Request System

The image shows a release selection interface where users request a software release instance by selecting a version and clicking "Request."

Actors

1. User – Selects and requests a release.
2. System – Processes the request.
3. Admin (Optional) – Manages releases. **Use Cases**
 - Select Release – User chooses a version (e.g., *Washington DC*).
 - Request Instance – User submits a request, and the system provisions the instance.
 - View Release Notes – User accesses release documentation.
 - Cancel Request – User cancels before submission.

1.2 Key Functionalities of the ServiceNow Virtual Agent

1. Automated Incident Management

- Users can report IT issues such as system crashes, software errors, or network failures directly through the chatbot.
- The Virtual Agent categorizes, prioritizes, and assigns incidents automatically.
- It can provide real-time updates on incident status

2. Service Request Automation

- Employees can request IT and business services, such as software installation, access provisioning, or hardware replacement.
- The Virtual Agent automates approvals and workflows, reducing manual processing time.

3. Natural Language Understanding(NLU) for Smart Interactions

- The chatbot understands user queries using AI-driven NLU models.

- It supports conversational interactions and recognizes intent, context, and sentiment for a human-like experience.

4. Knowledge Base Integration

- The Virtual Agent provides instant answers by retrieving relevant articles from the ServiceNow Knowledge Base.
- Users receive self-service solutions for troubleshooting common issues like password resets or software errors.

5. Workflow Automation and Integration

- The system integrates with **ServiceNow's backend** to automate tasks such as ticket creation, approvals, and escalations.
- It connects with third-party applications (e.g., Microsoft Teams, Slack) for **seamless communication**.

6. Customizable Conversation Flows

- Administrators can create and modify chatbot workflows tailored to specific organizational needs.
- The Virtual Agent can escalate complex issues to live agents when necessary.

7. Real-Time User Assistance

- Provides step-by-step troubleshooting for technical issues.
- Supports interactive forms where users can input required details for automated service processing.

8. Multi-Channel Support

- Available across various platforms such as web portals, mobile apps, Microsoft Teams, and Slack.
- Ensures accessibility across different devices for a seamless support experience.

9. Performance Monitoring & Analytics

- Tracks chatbot interactions and user satisfaction to measure effectiveness.
- Uses AI-driven insights to improve response accuracy and optimize workflows.

10. Scalability and Security

- Designed for enterprise-level scalability, handling thousands of interactions simultaneously.
- Implements role-based access control (RBAC) to ensure data security and compliance.

1.3 ServiceNow User Interface (UI)

The **ServiceNow User Interface (UI)** serves as the primary interaction point between users and the ServiceNow platform. Designed to be user-friendly, responsive, and highly customizable, the UI allows users across various roles—from administrators to end-users—to navigate, access, and manage enterprise workflows efficiently. The UI adheres to modern design standards and supports extensive configuration to match specific business needs, branding guidelines, and operational requirements.

ServiceNow's default interface includes a set of standardized components that facilitate smooth and intuitive interaction. These components are consistent across different modules and applications within the platform, which ensures a cohesive user experience throughout the system.

Below are the core components of the ServiceNow UI:

- Banner Frame
- Application Navigator
- SideBar
- Content Frame

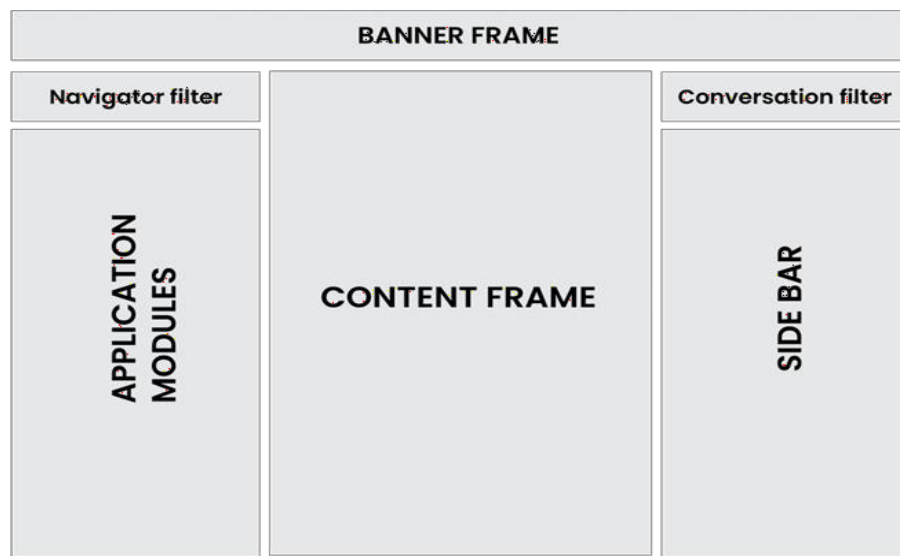


Fig 1.3 User interface overview

Following are the components of the user interface:

Application Navigator:

Application Navigator is a vital component that provides structured access to all available applications and their respective modules. It is typically located on the left-hand side of the screen and serves as a menu tree.

- Users can browse through different applications such as Incident, Change, Problem, Service Catalog, and many others.
- It contains a navigation filter at the top where users can type keywords or module names to quickly locate specific records or functions.
- As users type, ServiceNow dynamically displays relevant results, simplifying access and improving workflow efficiency.
- Administrators can control which applications and modules appear in the navigator based on roles and access permissions.

Banner Frame

The banner frame contains setting options, logo, user profile, option, and global search bar to switch between help and conversation in the right sidebar.

- It includes elements such as the **company logo, user profile options, settings,**

and the **global search bar**.

- The **user profile menu** offers quick access to preferences, profile settings, and logout options.
- The **global search bar** is a powerful tool that allows users to search across records, tables, and modules using keywords or record identifiers.

Side Bar

The **Sidebar** is a context-sensitive area that is not displayed by default. Users must activate it using options available in the banner frame.

The sidebar provides two main features:

- **Help:** When selected, the Help pane opens a panel that offers guidance and resources such as articles, training documents, and admin-published tips. This feature is useful for onboarding new users or offering support within specific modules.
- **Connect Chat (Conversation):** This feature allows users to engage in real-time messaging with colleagues, support teams, or administrators. It also retains **chat history**, which enables users to revisit previous conversations, track follow-ups, or refer back to shared information.

Content frame

The content frame is the primary part of the ServiceNow portal that records the complete content of the modules and applications. This frame also contains the main menu of the applications, which are essential from the administration perspective.

CHAPTER 2

VIRTUALAGENT OVERVIEW

2.1 What Is Virtual Agent ?

When users need help or to make a request, the interaction often takes the form of filling out a form, calling a service desk, or live chatting with a support agent.



Fig 2.1 Virtual Agent

Virtual Agent is a conversational bot platform that provides user assistance through conversations in a messaging interface.

Virtual Agent is always available to handle common requests and tasks, so users can receive help immediately. Configure Virtual Agent to run on platforms familiar to your users, such as third-party messaging apps, so users can interact with ServiceNow wherever they are working. Virtual Agent offers personalized customer experiences by applying and remembering user information during the conversation.

Typical support tasks that can be accomplished with Virtual Agent include:

- Answering Frequently Asked Questions (FAQ)
- Providing tutorial information
- Providing quick solution
- Reducing workload from the agent
- Gathering data for a live agent

2.2 Use Cases

The **ServiceNow Virtual Agent** revolutionizes enterprise support by automating key functions:

- **IT Support:** Instantly resolves technical issues, automates ticket creation, and streamlines incident management.
- **HR Services:** Assists employees with leave requests, payroll inquiries, and onboarding processes, reducing HR workload.
- **Business Support:** Automates approvals, optimizes workflows, and enhances decision-making with real-time insights.
- **Customer Service:** Provides 24/7 assistance, reduces wait times, and intelligently escalates complex issues for faster resolution.

2.3 Virtual Agent Environments

Runtime Environment	The Virtual Agent runtime environment is the running system responsible for user interaction. The behaviour of the Virtual Agent is based on the published topic, and the dialog interactions flow within that topic. On the NLU side, the topic returned to the user is based on the published and trained NLU model and the user's utterance.
Design Environment	The Virtual Agent design environment is configured in separate interfaces. This allows the administrator to make modifications to either the conversation flow and entity/intent processing (in the Virtual Agent Designer) or the trained NLU model (in the NLU Workbench). The changes made in the design environment only show up in the runtime environment after publication.

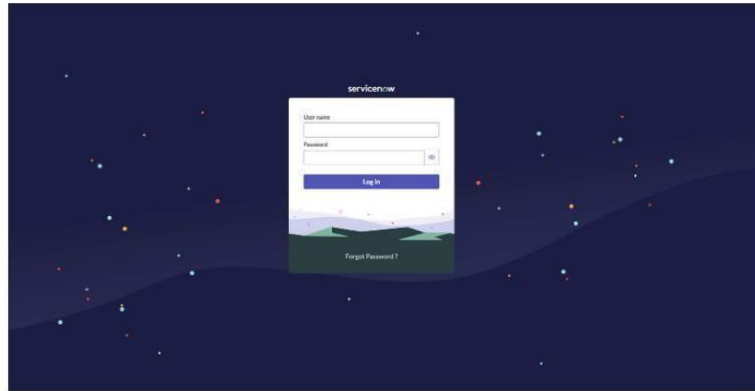
Table 1: Virtual Agent Environments

CHAPTER 3

SETTING UP VIRTUALAGENT

3.1 Logging Into The Developer Instance

An instance



An instance in the Platform is independent, changeable, and highly configurable. If you can dream it, you can do it!

Fig 3.1 Signup Page

Developer Website: developer.servicenow.com

login by using the username and password the developer interface will be shown as below:

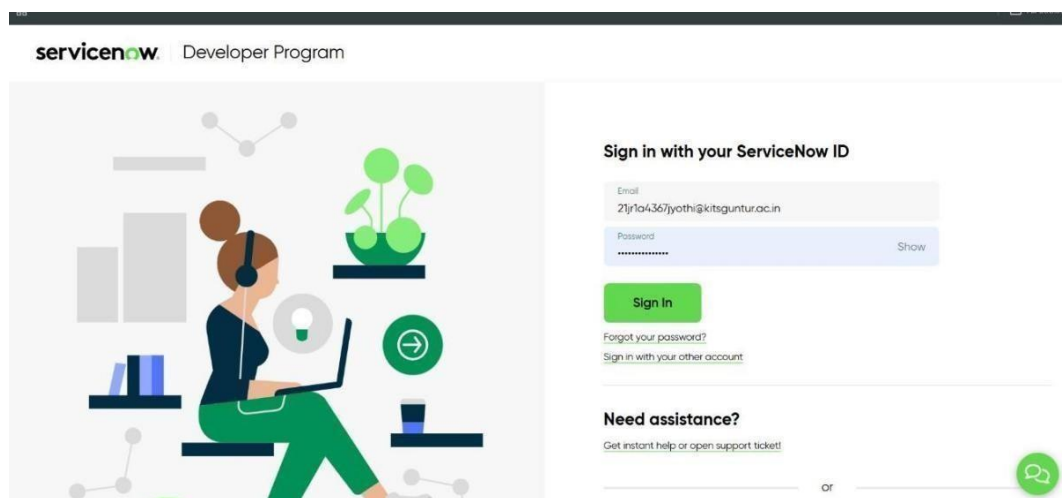


Fig 3.2 Login page

A Personal Developer Instance (PDI) in ServiceNow is a free sandbox environment provided to members of the ServiceNow Developer Program. It allows developers to learn, explore, and experiment with the Now Platform without impacting a customer.

Free Access:

PDIs are available for free to Developer Program members.

After signup with your credentials the page will be open as

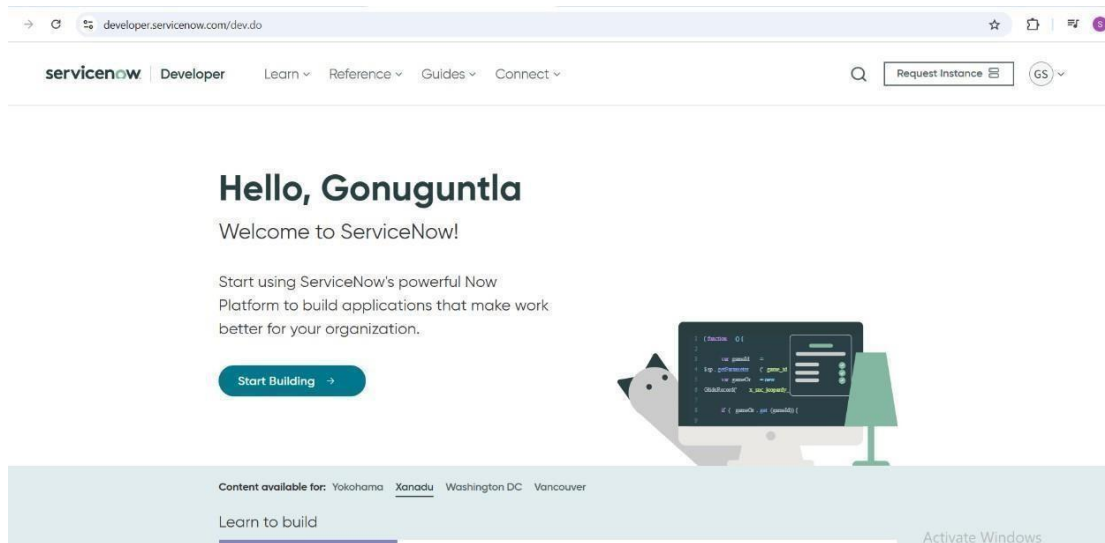


Fig 3.3 Home page for Instance

By clicking on the “**Start Building**” the interface is shown below and it is ready to develop tasks.

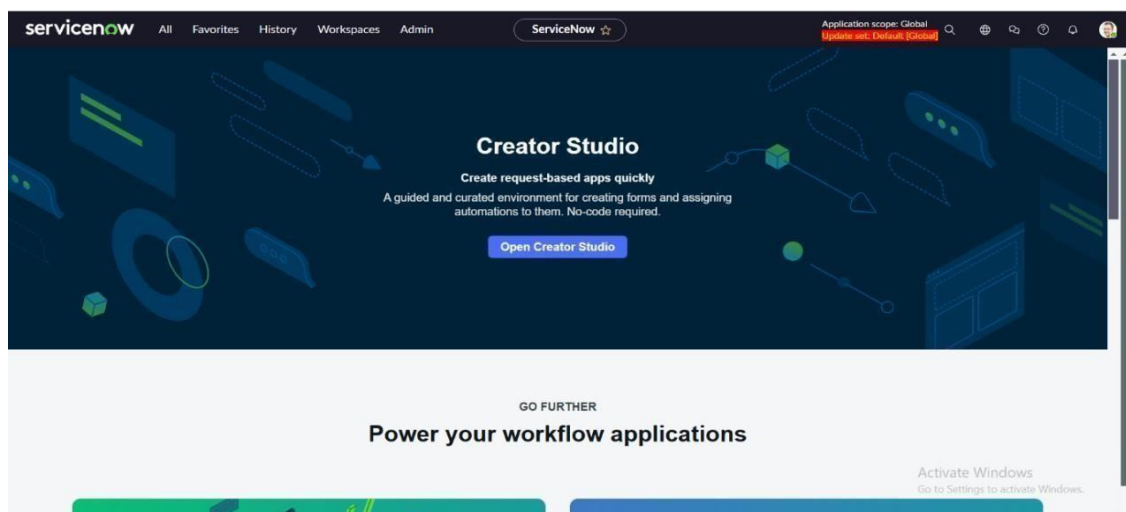


Fig 3.4 Home page

Use Creator Studio to build request-based applications without writing code.

Streamline Your Workflow

Power your operations with tailored workflow solution

3.2 Go to all Navigator and Search for Conversational Interfaces Guided Setup.

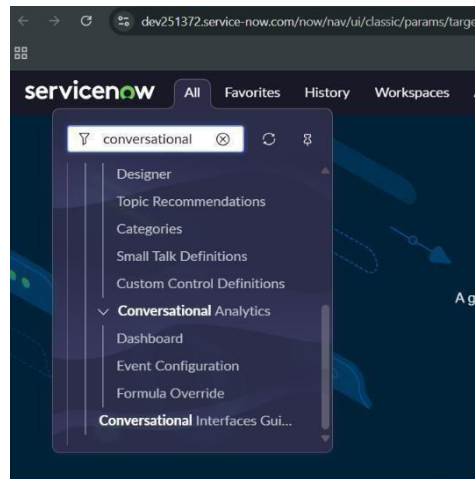


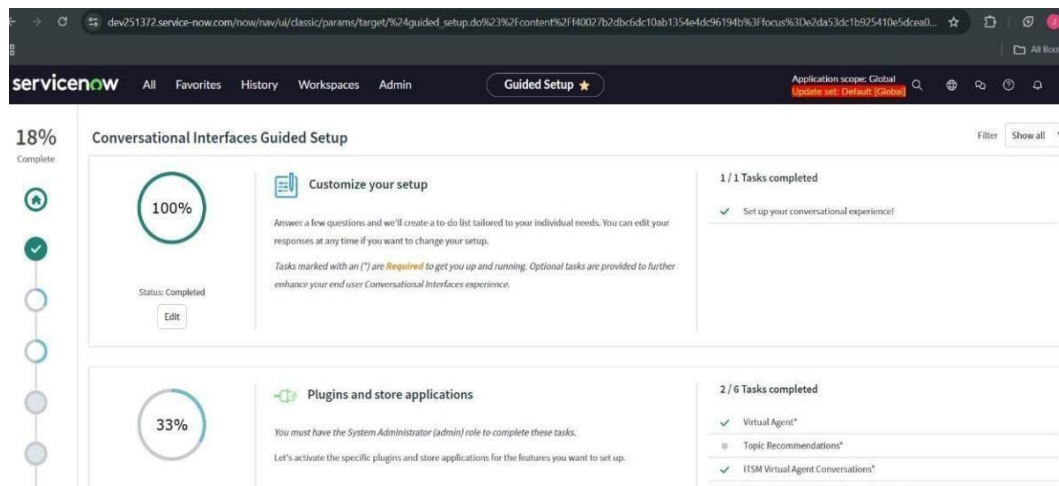
Fig 3.5 Application Navigator

This Conversational Interfaces Guided Setup consists of

1.  **Customize your setup:** Answer a few questions and we'll create a to-do list tailored to your individual needs. You can edit your responses at any time if you want to change your setup.
2.  **Plugins and store applications:** You must have the System Administrator (admin) role to complete these tasks. Let's activate the specific plugins and store applications for the features you want to set up.
3.  **Client configuration:** Configure and customize the Service Portal web client, branding, Now Mobile application, and other chat experiences to meet the specific needs of your company.
4.  **Virtual Agent:** Explore and define Virtual Agent content to provide your end users with intelligent, personalized, meaningful conversations that actually get things done.

5.  **Surveys:** Conversational Interfaces supports surveys in two different forms: Pre-Chat Surveys & Post-Chat Surveys. You can use Pre-Chat Surveys to get more context and concisely route chats to the right agents. You can also use Post-Chat Surveys to collect end user feedback on Virtual Agent or Workspace Agent Chat conversations.
6.  **Pre-built integrations:** Install and configure pre-built integrations for Virtual Agent and Agent Chat to provide your end users with the help they need within Microsoft Teams, Slack, Workspace from Facebook, SMS with Twilio, and Bot-to-bot.
7.  **Custom integrations:** Sometimes one size doesn't fit all. If the pre-built integrations aren't right for your organization, you can build custom integrations (including new messaging channels).
Configure custom integrations to a channel of your choice and bring Virtual Agent and/or Workspace Agent Chat to your end users. There are two integration options available.
8.  **Conversational messaging:** Conversational messaging channels allow Workspace agents to support long-running conversations with end users. Unlike standard chat channels, these service channels can support much longer timeouts than chat. The default pre-built channel is SMS with Twilio.
9.  **Notifications:** Enable the system to send chat notifications directly to your end users in the web client, Slack, Microsoft Teams, and Workplace from Facebook for both Workspace Agent Chat and Virtual Agent.
10.  **Dashboards:** Explore pre-built analytics dashboards that enable you to monitor user engagement with Virtual Agent conversations and metrics related

to Agent Chat.



- Click on Customize your setup

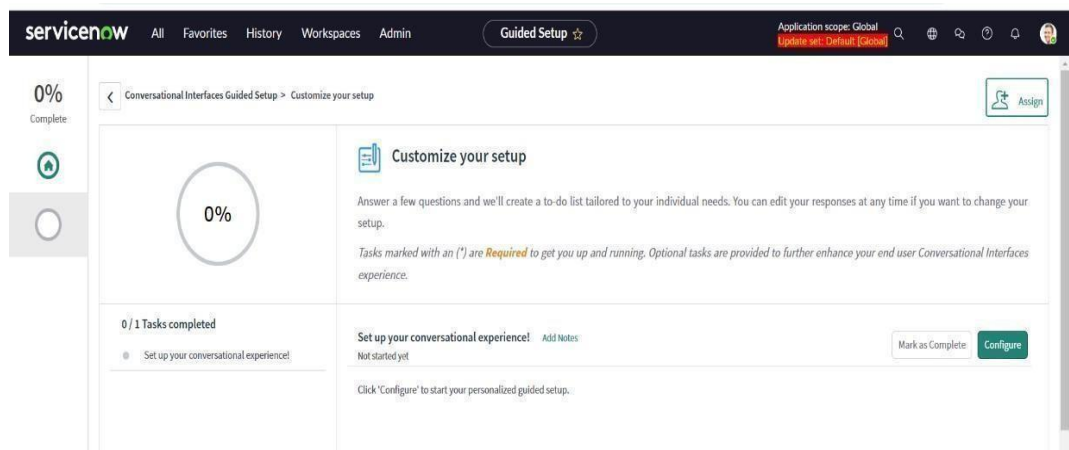
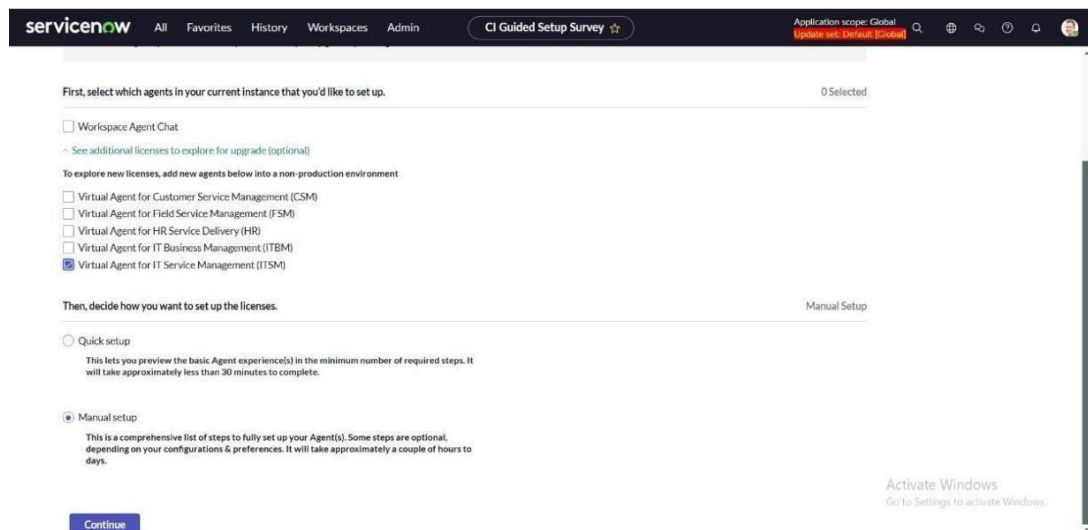


Fig 3.6 Conversational Guided Setup

- Then click on “**Configure**”.
- After click on the Configure the different types of agents are available to set up.
- Select any one of the from the different types of agents.
- Select the Virtual Agent for IT Service Management(ITSM).
- And select the manual setup.



Click on “Continue”.

3.3 To Activate Virtual Agent we need to install Plugins

Click “Get Started” on “Plugins and Store Applications”.

- To enable Virtual Agent, activate the Glide Virtual Agent plugin (com.glide.cs.chatbot).

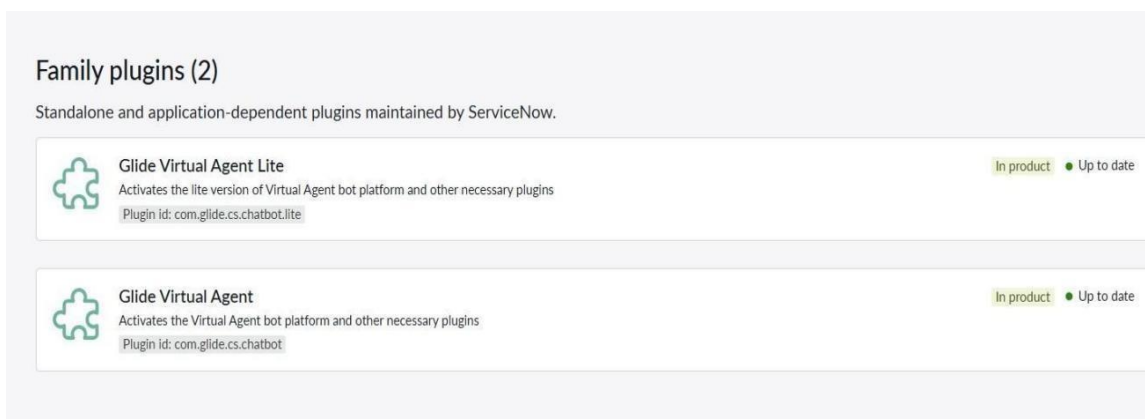


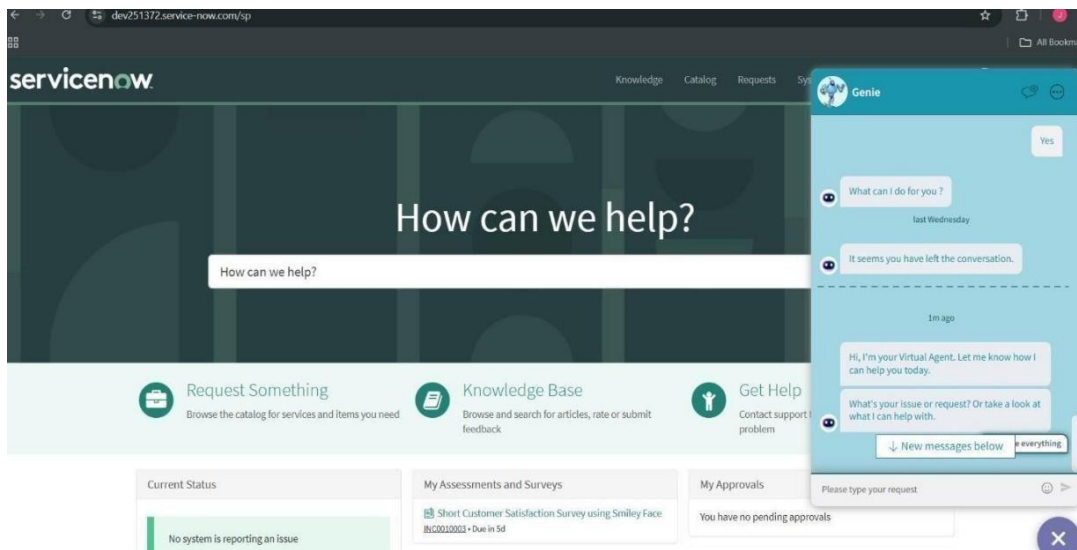
Fig 3.7 Plugins

- After Installing Plugins
- Login to Service Portal (sp)

Different way to move to service portal :

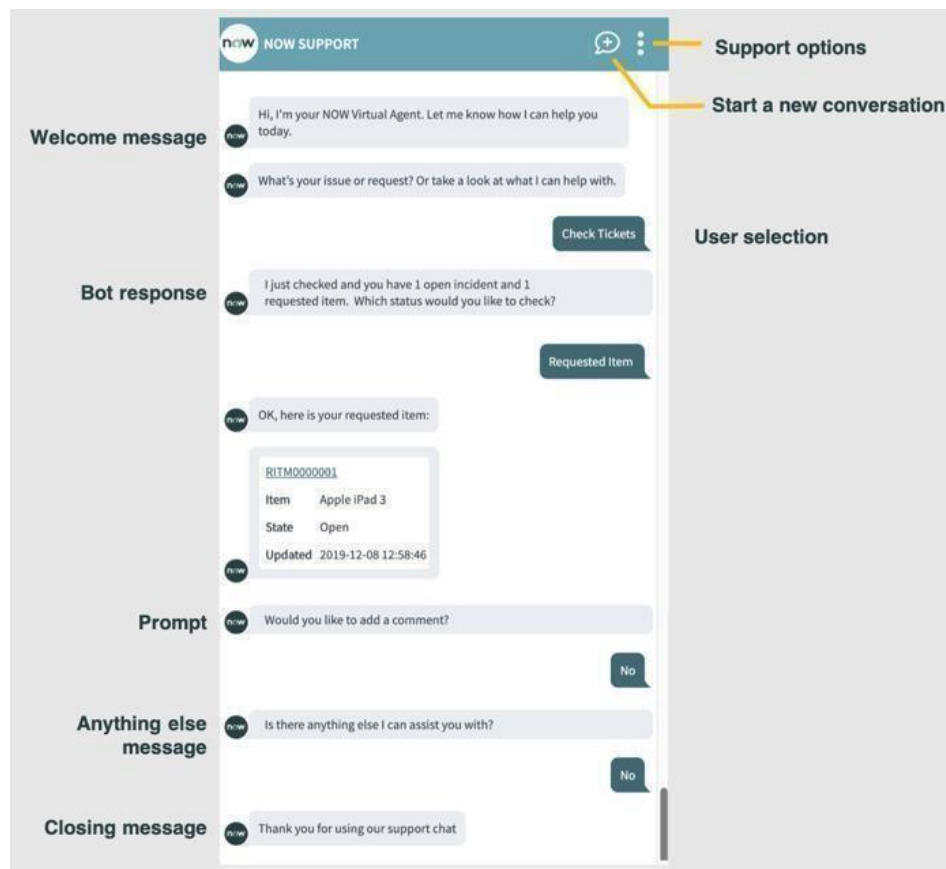
- [https://dev251372.service-now.com/\\$sn-va-web-client-app.do](https://dev251372.service-now.com/$sn-va-web-client-app.do)

- <https://dev251372.service-now.com/sp>



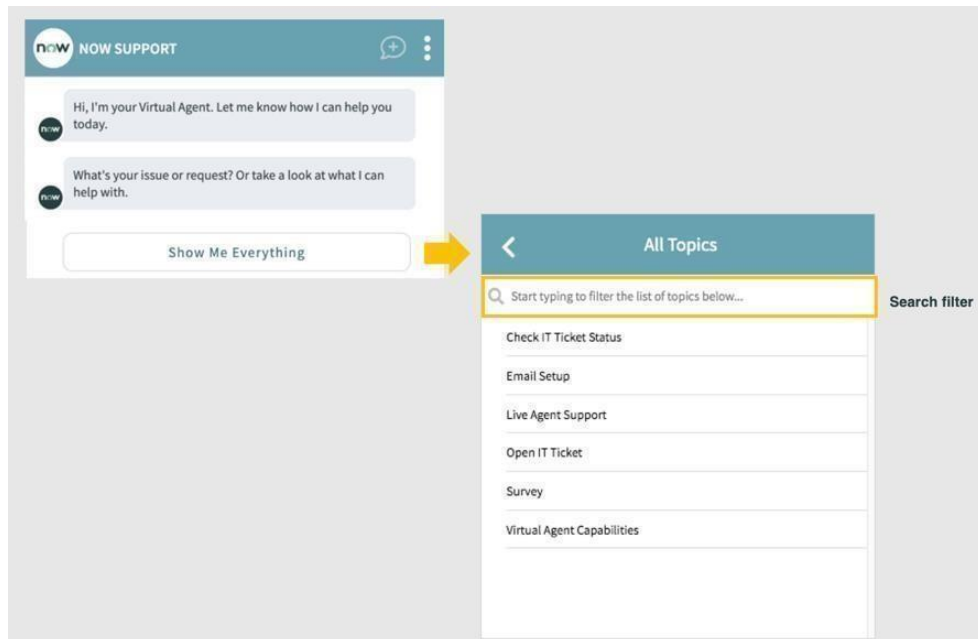
3.4 Conversational Interface

The Virtual Agent conversational interface facilitates an automated messaging session between the user and the virtual agent. The web client interface makes use of the Service Portal web widget to configure for custom service portals. The device is available in the CSM, [HR Service Delivery](#), and [ITSM Service](#) Portals in ServiceNow.



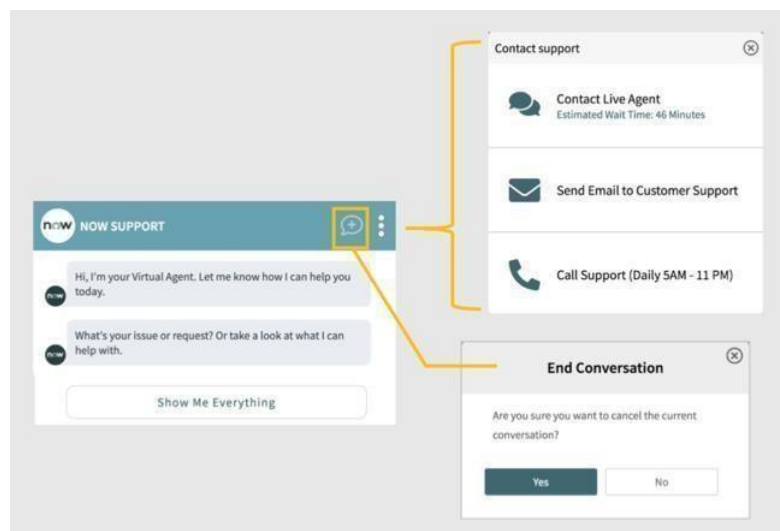
3.5 Conversation Options

When users start a conversation with the bot, they can enter a request or see a list of everything the bot can help. First, the chat window flashes all topics accessible to the user. Then, users use the search box and filter the list of issues if they choose everything.



3.6 Manage the Conversation

The conversational interface offers users several options to manage the conversation. For example, users can stop the current discussion and initiate customer support to access a live agent for immediate assistance.



CHAPTER 4

CUSTOMIZING VIRTUALAGENT

4.1 Branding Virtual Agent

Click “Get Started” on “Client Configuration”.

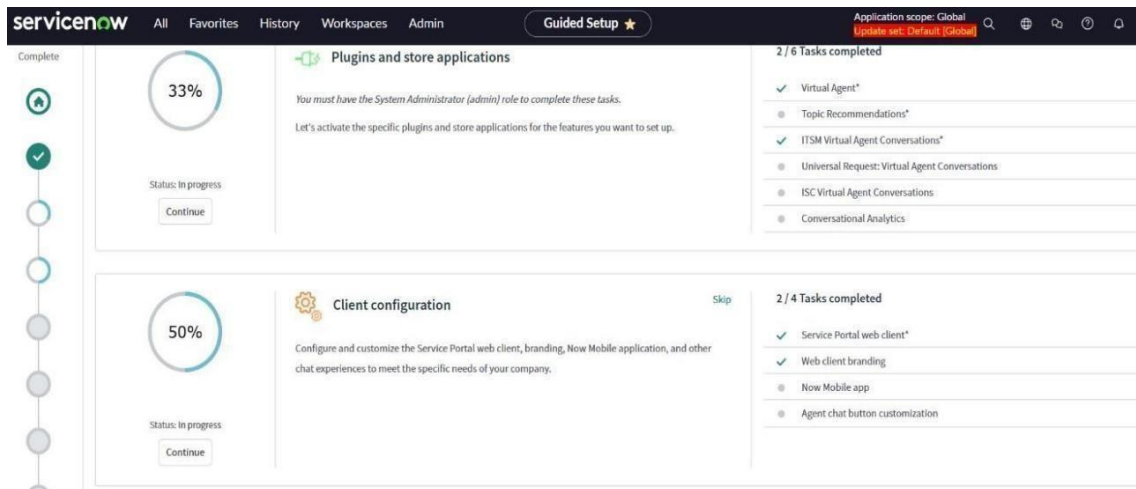
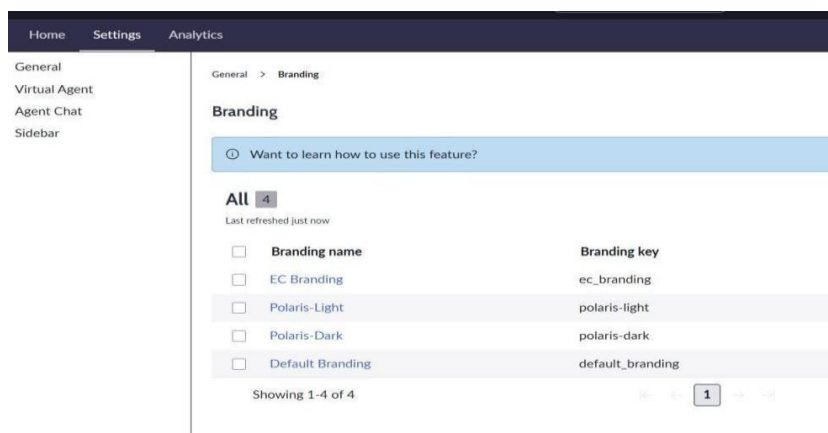
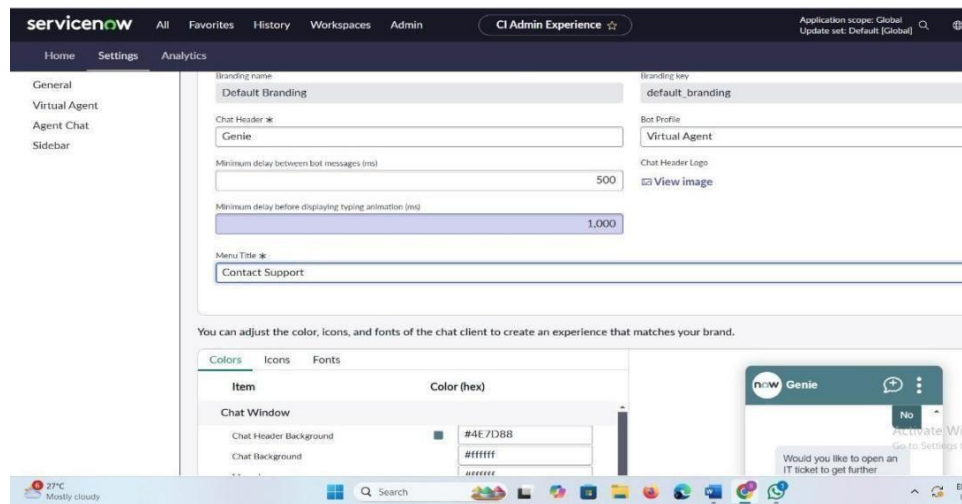


Fig 4.1 Branding page

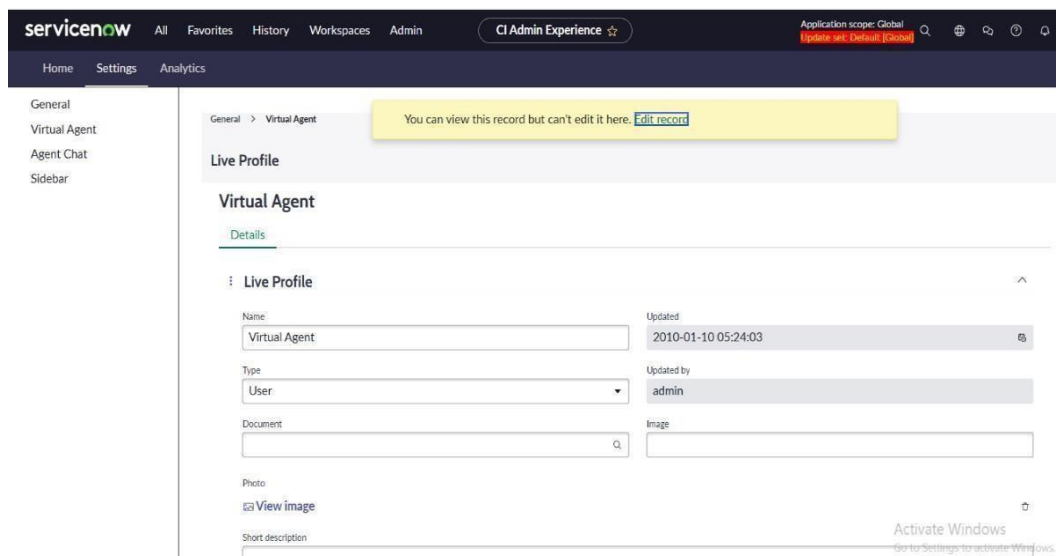
- **Branding of a virtual agent:** Customize the look and feel of your web client to better represent your organization. You can create different styles and experiences for multiple organizations within your instance, also referred to as ‘Multi-Branding’.
- The branding of a virtual agent involves in
 - I. Name
 - II. Logo
 - III. Chatting Interface
- **Changing the name:** Go to Web client branding and click on “Configure”.



- And select the “**Default Branding**”. And change the Chat Header with required name.



- **Changing the name:** Change name according to you
Here as “**Genie**”.
- **Changing the Bot Profile:** Default as Virtual Agent



clicking in “**Edit Record**” the record is to be updated that we can made changes to the logo

Live Profile Virtual Agent

Name: Virtual Agent

Type: User

Document: [Search Icon] [Add Icon]

Photo: [Update] [Delete]

Updated: 2025-03-11 08:18:18

Updated by: admin

Image: [Image Icon]

Short description: [Text Field]

[Update] [Delete]

- Update and save the changes.
- Then go back to service portal and see the changes.

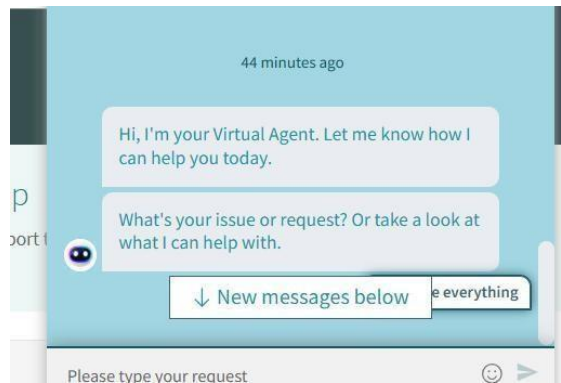
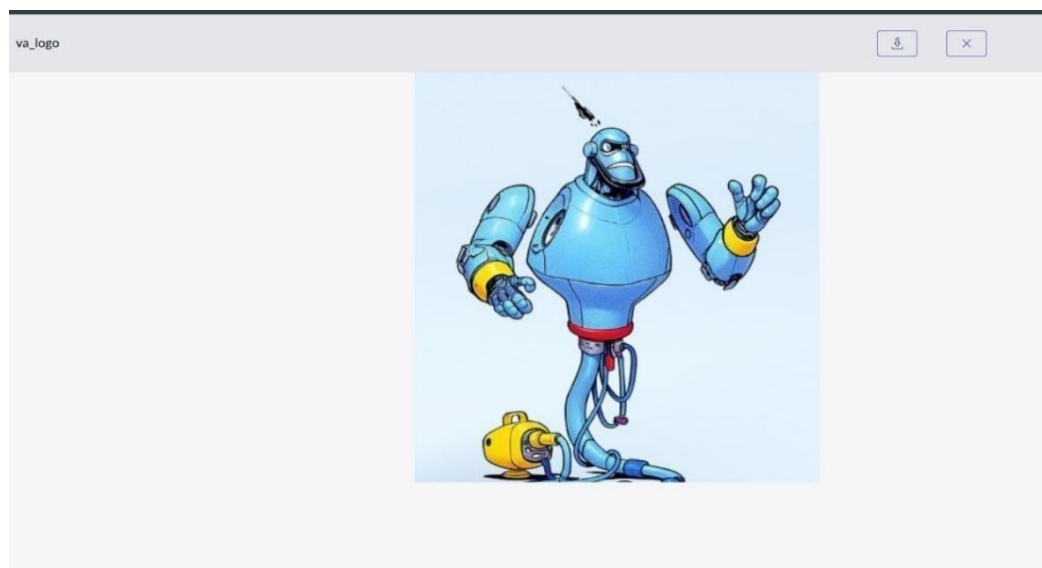
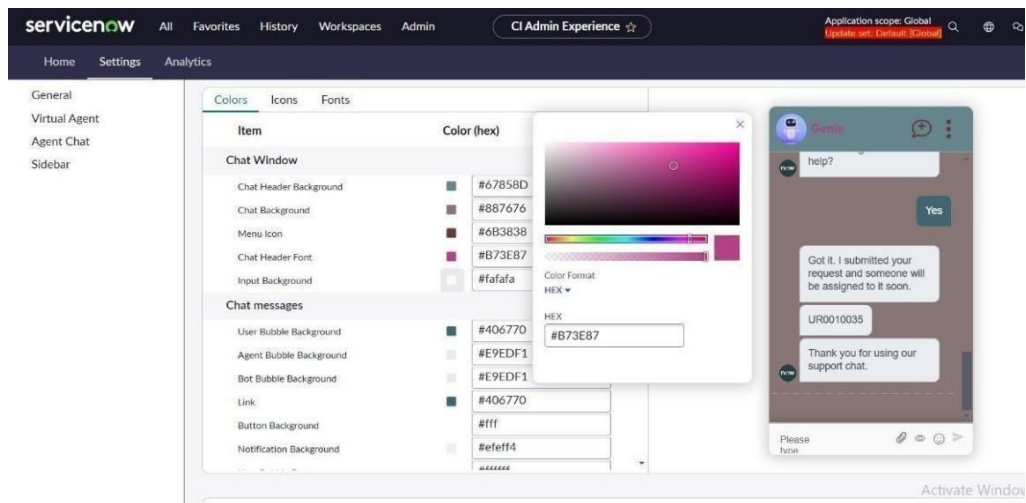


Fig 4.2 Service portal screen

- You can Change the “Chat Header logo”.

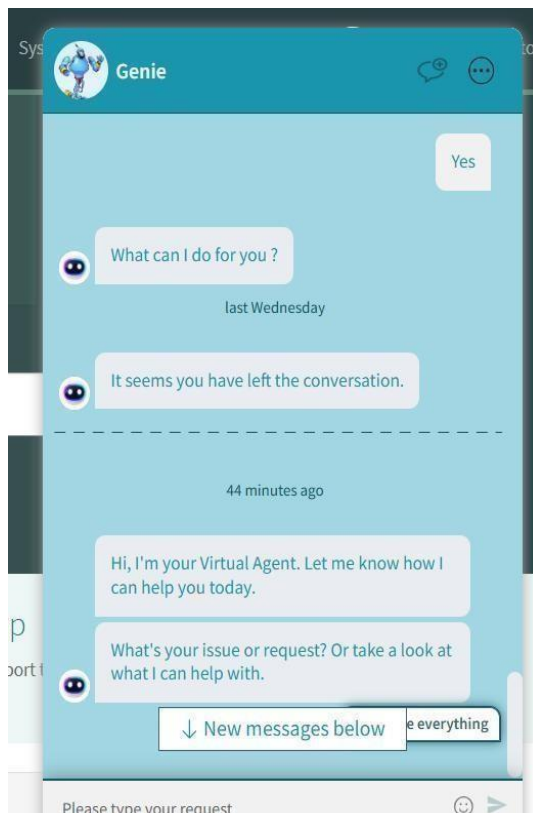


- You can adjust the color, icons, and fonts of the chat client to create an experience that matches your brand.



- We can also change **Minimum delay between bot messages (ms)** and **Minimum delay before displaying typing animation (ms)**.
- After Customizing our virtual agent with the requirements it appears as below.

Our customized virtual agent:



Before Customized:

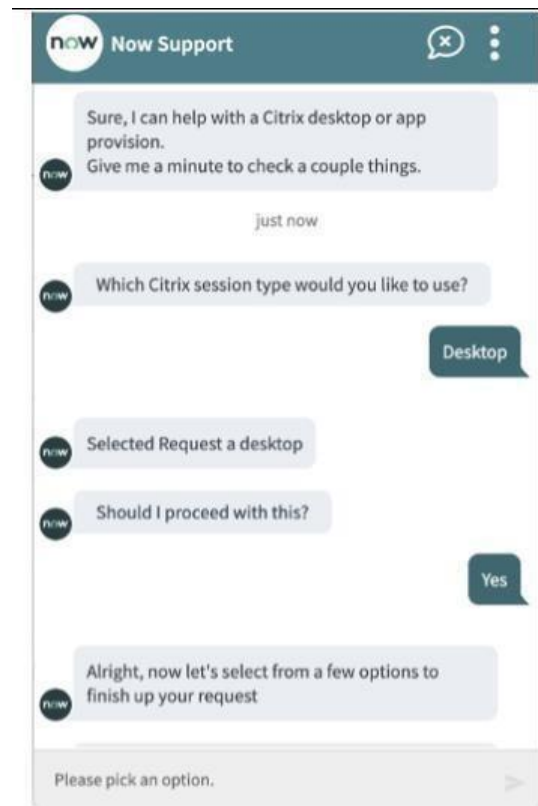


Fig 4.3 Customizations

4.2 Creating Custom Greetings

Greeting: The greeting message that come from the bot(or)virtual agent.

Custom greetings help define how a bot interacts with users at the beginning of a conversation. A well-crafted greeting enhances user engagement and sets expectations for the conversation. To create and configure a custom greeting, follow these steps:

Steps to Configure Custom Greetings

1. Access the Configuration Page

- After activating the Virtual Agent or Virtual Agent Lite, navigate to the settings menu.
- Open the configuration page where bot settings and interactions are managed.

2. Setup Topics for Custom Greetings

- Since we are customizing greetings, we need to define a **topic** that dictates how the bot should initiate a conversation.
- Topics help structure the flow of interaction between the bot and users.

3. Select the Setup Topic as “Dynamic Greeting Topic”

- Choose “**Dynamic Greeting Topic**” as the setup topic.
- This topic allows the bot to deliver greetings that adapt based on user behavior, previous interactions, or system conditions.
- Save the record to apply the changes.

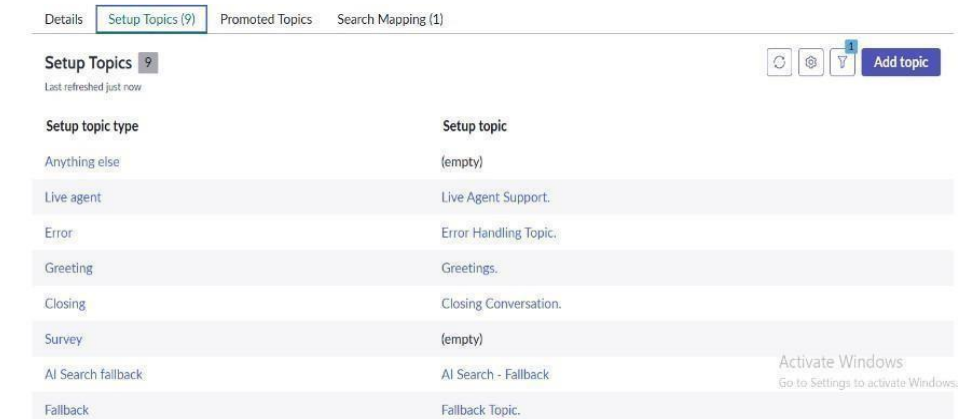
What is a Dynamic Greeting Topic?

A **Dynamic Greeting Topic** is a predefined logic that determines how a virtual agent greets users based on various conditions. Instead of using a static greeting (e.g., "Hello! How can I help you?"), a dynamic greeting adjusts its message based on:

- **User’s previous interactions** (e.g., "Welcome back! How can I assist you today?")
- **Time of day** (e.g., "Good morning! Need help with anything?")

- **User’s profile data** (e.g., "Hi [User's Name], do you need support with your last request?")
- **Bot’s current state** (e.g., "System updates are in progress. How can I assist you?")

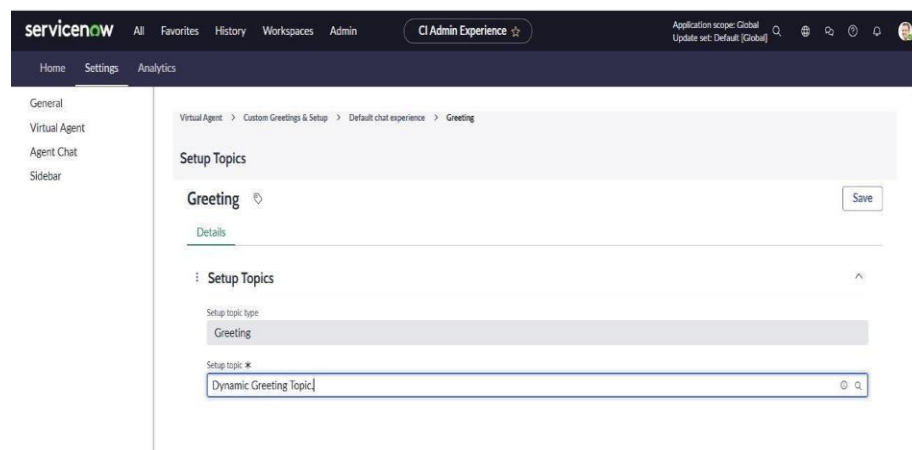
To create the custom greetings click on the “**Custom Greetings & Setup**” it is required to activate the plugin



Setup topic type	Setup topic
Anything else	(empty)
Live agent	Live Agent Support.
Error	Error Handling Topic.
Greeting	Greetings.
Closing	Closing Conversation.
Survey	(empty)
AI Search fallback	AI Search - Fallback
Fallback	Fallback Topic.

Fig 4.4 Custom Greeting setup

- Activation of either the Virtual Agent or Virtual Agent Lite plugin is required to access the configuration page for this task.
- As we are creating customizing greetings we are going to setup topics
- Select the setup topic as “**Dynamic Greeting Topic**”. And save the record.
- **Dynamic Greeting Topic:** This dynamic greeting bot is defined as how the bot will behave if you start the bot.



Virtual Agent > Custom Greetings & Setup > Default chat experience > Greeting

Setup Topics

Greeting Save

Details

Setup Topics

Setup topic type: Greeting

Setup topic: Dynamic Greeting Topic

- Now, go to the service portal, then the agent shows the greetings as follows

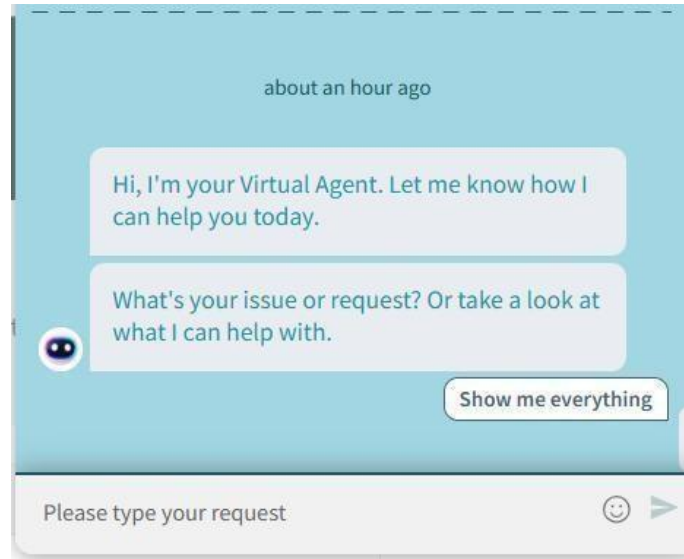
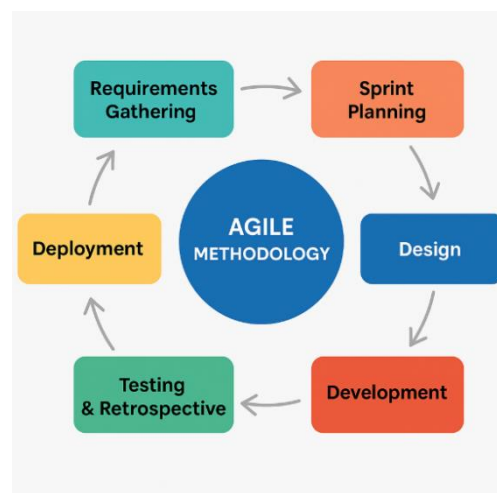


Fig 4.5 Greetings page

Creating custom greetings using the **Dynamic Greeting Topic** enhances the way virtual agents interact with users by making conversations more personalized and context-aware. By leveraging user behavior, time of day, and system conditions, dynamic greetings provide a more engaging and efficient user experience. Once configured and activated through the Virtual Agent or Virtual Agent Lite, these greetings can be seamlessly displayed on the Service Portal, ensuring that users are welcomed in a thoughtful and relevant manner every time they interact with the bot.

Agile Methodology



1. Requirements Gathering

For your Virtual Agent project, this phase involves:

- Identifying common IT and business support queries (e.g., password resets, incident reporting).
- Collaborating with stakeholders to understand user pain points.
- Gathering initial conversation topics and intents for the chatbot.
- Reviewing existing ServiceNow functionalities and integrations.

Outcome: A backlog of user needs that the Virtual Agent must address.

2. Sprint Planning

This is where your team defines what features to build in each sprint:

- Select key functionalities like incident creation, service requests, or FAQ responses.
- Break larger tasks like NLU integration or multi-platform support into smaller sprint-sized chunks.
- Assign priorities based on urgency and user value.

Example Sprint Goal: Implement basic ticket-raising conversation flow.

3. Design

In this phase, the conversational experience and system architecture are defined:

- Design chatbot conversation flows (e.g., greeting > issue selection > details > confirmation).
- Plan UI enhancements in the ServiceNow portal.
- Structure NLU training data to recognize varied user input.
- Design data flows for integration with ITSM modules.

Tools: ServiceNow's Virtual Agent Designer, flow diagrams, mock interfaces.

4. Development

Your team begins building the Virtual Agent in ServiceNow:

- Create topics and dialogs in Virtual Agent Designer.
- Set up integrations with backend systems like incident management and knowledge base.
- Implement fallback responses and escalation to live agents.

- Use JavaScript or ServiceNow scripting for advanced logic.

Result: A working conversational agent that can handle selected scenarios.

✅ 5. Testing & Retrospective

This phase ensures everything works before deployment and reflects on the sprint:

- Conduct unit testing for each chatbot topic and subflow.
- Test conversation accuracy using varied user inputs.
- Gather early user feedback and track analytics (e.g., conversation success rates).
- Hold a retrospective to discuss what went well and what needs improvement.

Goal: Refine the chatbot experience based on performance and feedback.

🚀 6. Deployment

Once testing is complete:

- Deploy the Virtual Agent to a staging or production environment in ServiceNow.
- Release new features in stages (e.g., IT helpdesk → HR support → Procurement).
- Monitor via performance analytics to identify improvement areas.
- Gather feedback through user surveys and conversation logs.

Goal: Continuously deliver improvements with each Agile sprint.

🔄 Agile Loop in Action

Your project repeats this cycle with each new sprint. For example:

- Sprint 1: Ticket creation and status checking
- Sprint 2: Knowledge base integration
- Sprint 3: HR queries support
- Sprint 4: Multilingual support

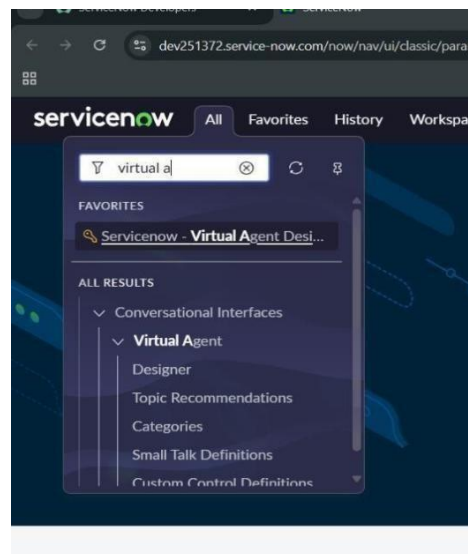
CHAPTER 5

BUILDING CONVERSATIONS & TOPICS

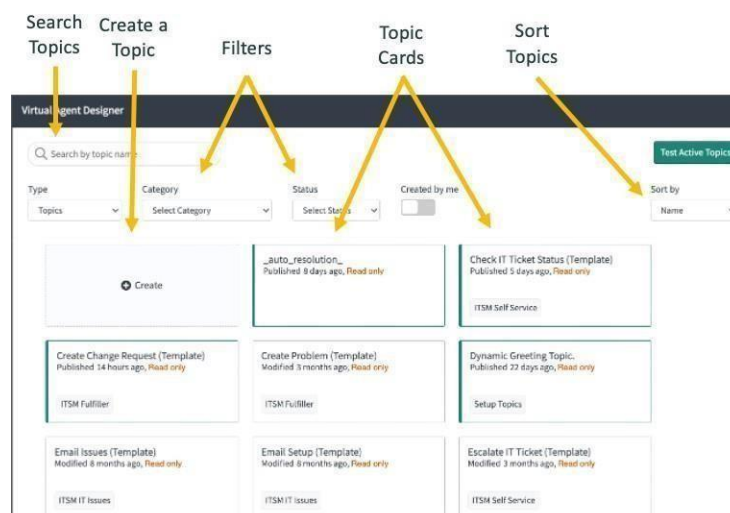
5.1 Creating A Topic

Topic: A set of instructions for the bot. The topics are created to accomplish a goal and build topic flows to describe the different paths a topic can take.

Go to all navigator and search for “**Virtual Agent > Designer**”.



- Click on create to **create**

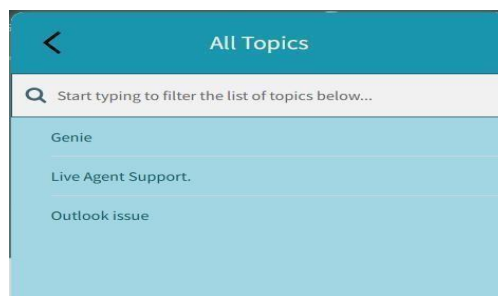


- The created topic is related to the “**outlook issue.**”
- Fill the mandatory details that are required to create the topic.
- In this topic the keywords related to the issue also be entered to encounter

- Under the categories select “ITSM IT Issues”.

5.1 ITSM Issues

- And click on create.
- **"Default start topic"**: It is a custom topic which the bot just starts with? The bot start with the default sentences and the Show Me Everything button.
- The all created topics are seen in agent under **Show Me Everything button**.
- The topics which are in the agent was shown in the figure.



5.2 Designing Topic Flows

Introduction

Designing topic flows in ServiceNow Virtual Agent ensures a structured and efficient conversation, improving user experience by guiding interactions logically.

Key Steps for Topic Flow Design

1. Define User Intent
 - Identify common user requests and categorize intents.
 - Use NLU (Natural Language Understanding) for better intent recognition.
2. Create and Configure Topics
 - In Virtual Agent Designer, select Create New Topic and provide a name & description.
 - Define trigger phrases to activate the topic (e.g., “reset password”).
3. Structure the Conversation Flow
 - Start with a greeting message and clarify the user’s request.
 - Use questions, choices, and decision trees to guide interactions.
 - Offer self-service options where applicable.
4. Apply Conditions and Branching
 - Use If-Else conditions to adjust responses based on user input.
 - Ensure smooth transitions between related topics.
5. Integrate with ServiceNow Actions
 - Automate actions such as creating incidents or fetching user details.
 - Use API calls and system records to enhance functionality.
6. Test and Optimize

- Use the Test Chat Interface to simulate interactions.
- Monitor conversation logs and improve based on feedback.

Drag a control from the palette to the canvas. Drop points (



) appear on the transition lines where the control can be added. Drag the control over a drop point to create a node. Drop points change color and are highlighted with a dotted line when the control can be dropped.

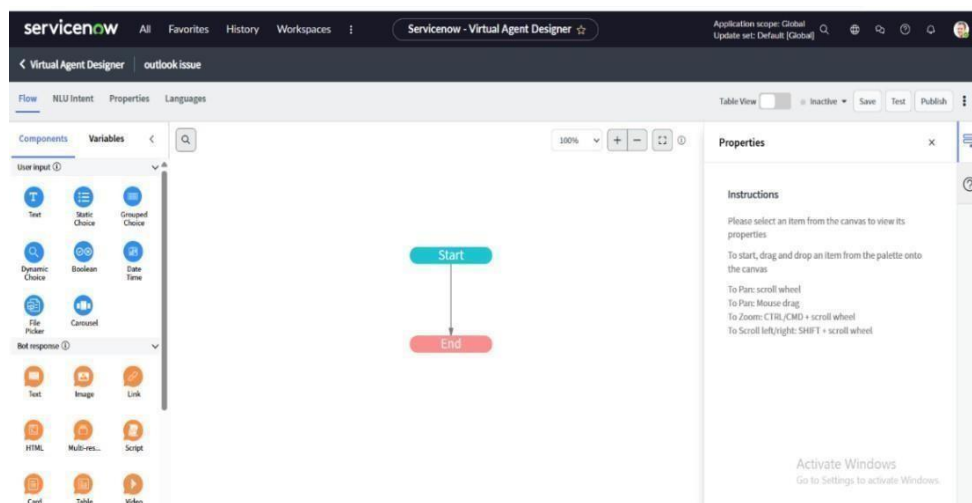
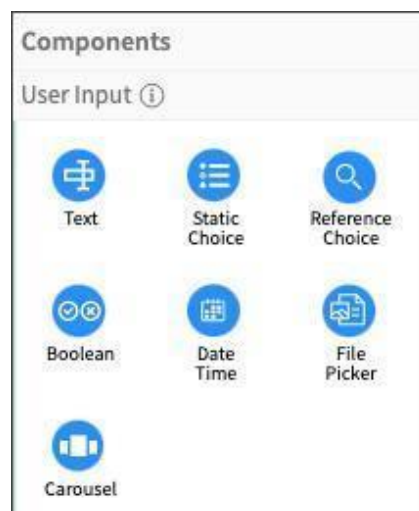


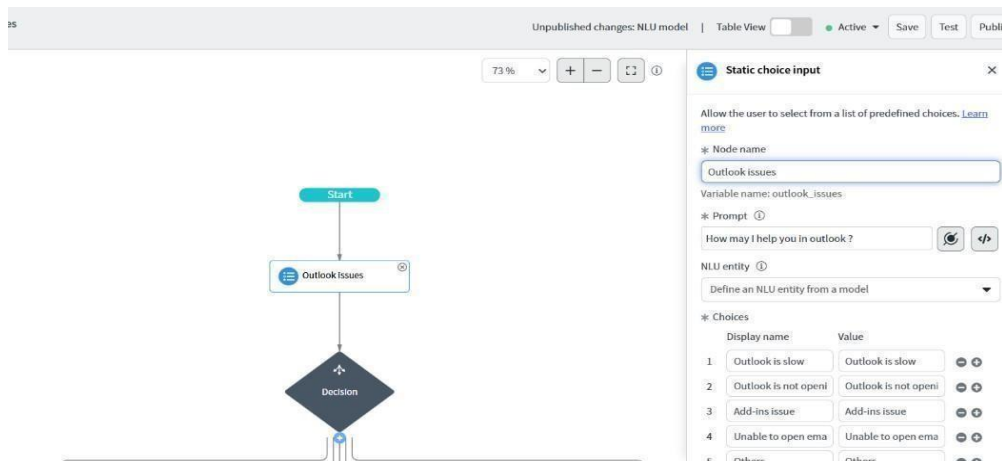
Fig 5.2 Virtual Agent Designer

Getting User Input

Add *User Input* controls to a topic flow to capture input from a user.



- The left side of the flow chart was in-built with different types of user inputs, bot response, utilities to build the topic flow.



5.2.1 Building the Topic Flow: Setting Up Greetings

The first step in designing a topic flow in ServiceNow Virtual Agent is configuring the **greeting message**. The greeting is essential as it initiates the conversation and sets the tone for user interaction.

Steps to Set Up the Greeting in the Topic Flow

1. Start the Topic Flow Design

- Open the **Virtual Agent Designer** in ServiceNow.
- Click on **Start** to begin creating a new topic flow.

2. Access the Greeting Tab

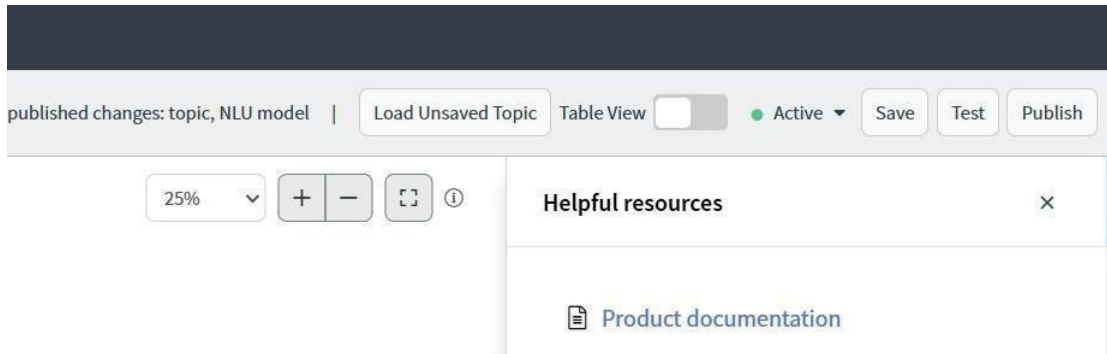
- Once inside the topic flow editor, locate and **click on the "Greeting" tab**.
- This section allows customization of the bot's initial message to users.

3. Enter the Greeting Message

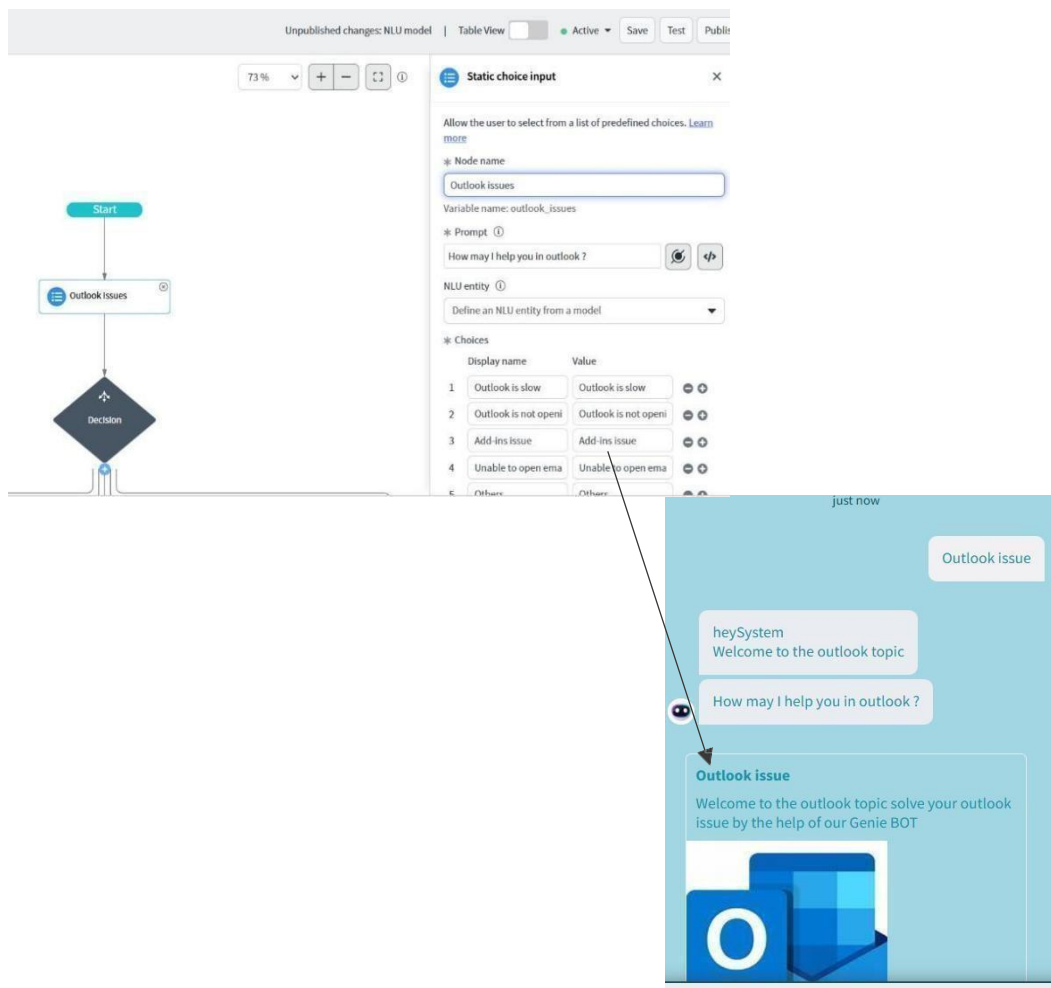
- In the greeting input field, type a welcome message such as:
"Hello System! Welcome to the Outlook topic."
- This message ensures a friendly and engaging start to the conversation.

4. Test the Greeting Message

- Click on the "Test" button to preview the greeting.
- This helps verify that the greeting appears as expected when a user initiates the conversation.



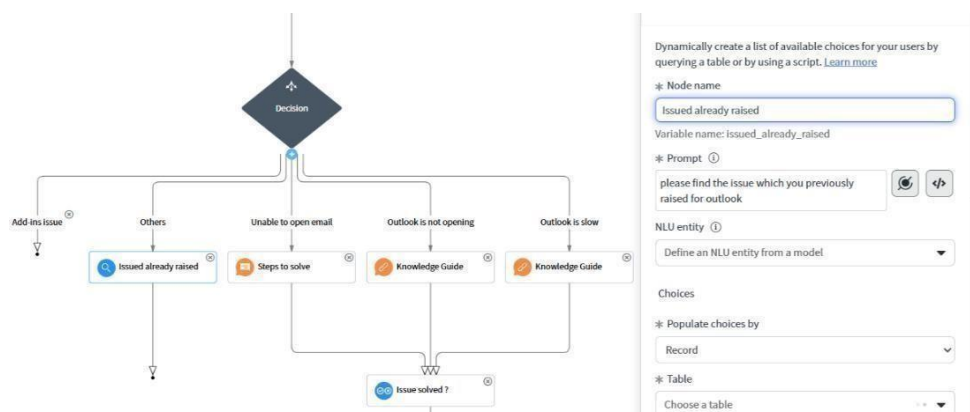
Then the output will be generated as:



- Start building the flow by adding the text to the flow
- After creating the flow, test the flow.



- Add a Decision element from the Utilities section to the flow to guide the next steps.
- Define multiple branches to handle different user responses or scenarios.
- Assign a name to each branch for clarity and better flow management.
- Set appropriate conditions for each branch to determine the conversation path.



5.2.2 Analysis of the Topic Flow

1. Decision Node

- The flow starts with a **Decision** node, branching into multiple troubleshooting paths based on user input.

2. Branching Options

- **Add-ins Issue:** Likely directs users to a solution for Outlook add-in problems.
- **Others:** Checks if the issue was already raised and prompts relevant information.
- **Unable to Open Email:** Leads to a **Steps to Solve** guide for troubleshooting.
- **Outlook Not Opening & Outlook is Slow:** Direct users to a **Knowledge Guide** for resolution.

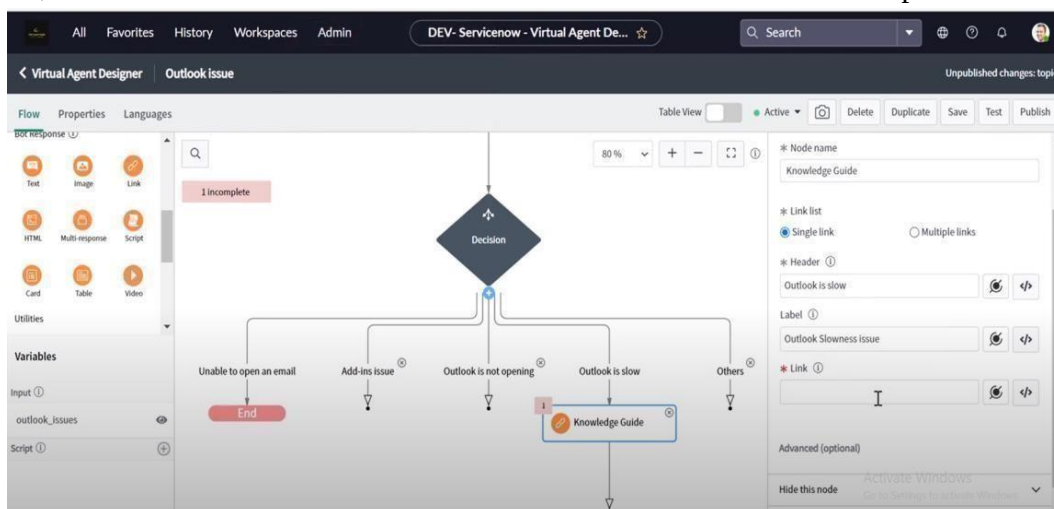
3. Issue Solved Check

- After guiding users through the process, a follow-up node asks: "**Issue Solved?**"
- This ensures that the user's problem is resolved or provides further steps if needed.

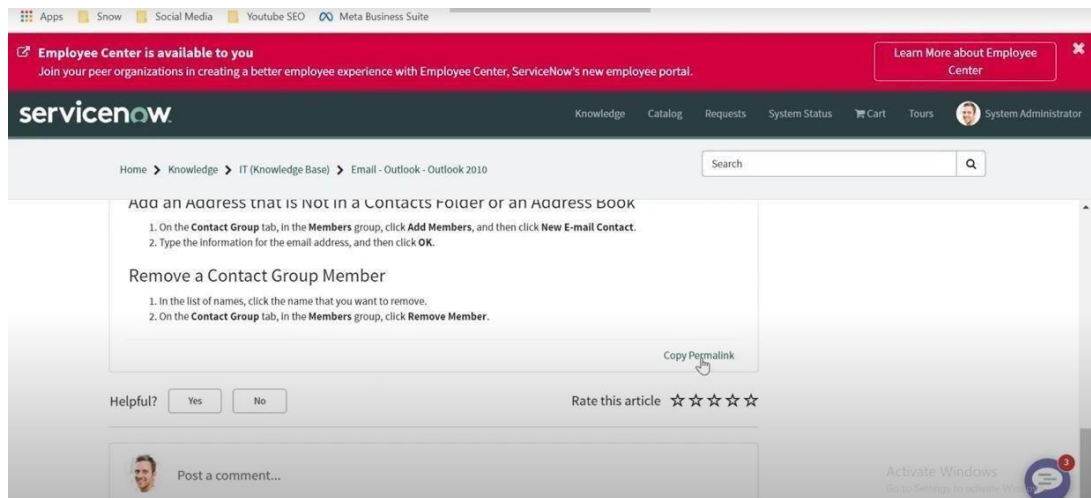
4. Dynamic Choice Population

- The right panel shows a dynamically generated list of choices based on a **record table** or a script query.
- Users are prompted to find previously raised issues related to Outlook.

Next, add a link to the flow which directs the user to the troubleshoot steps

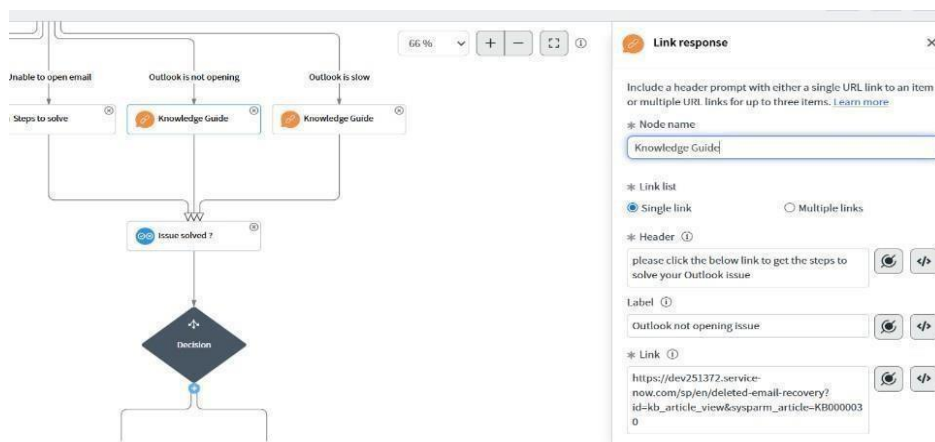


- To document the solution, navigate to the **Knowledge Article** section and create a new article detailing the steps to resolve the issue. Ensure the article includes clear instructions and troubleshooting guidance. Once the article is saved, note the assigned **Knowledge Base (KB) number**.
- Next, open the **Service Portal** and enter the KB number into the search bar. This will generate a "**Copy Permalink**", which can be used to easily share the article with users seeking a solution.



Click on copy permalink and paste link on the left side of the flow.

Paste in on the “Link”.



Boolean Input for User Confirmation

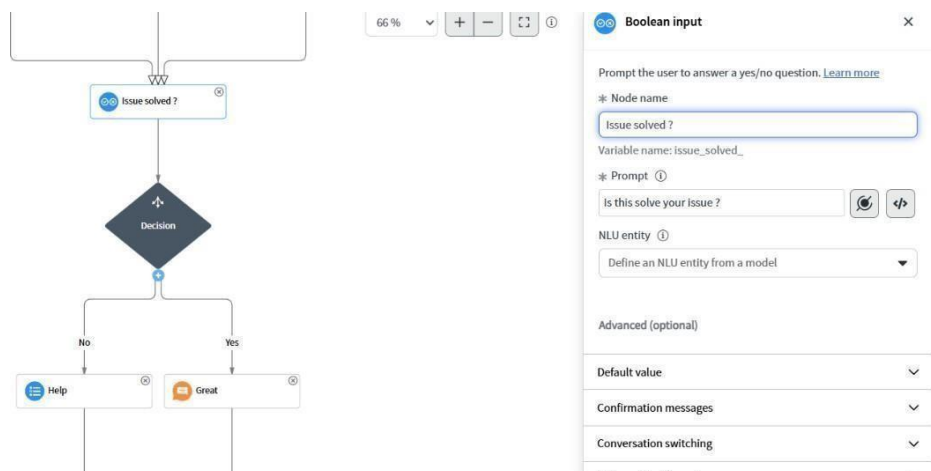
- The **"Issue solved?"** node asks users if their issue has been resolved.
- The **prompt message** ("Is this solving your issue?") ensures clarity in user responses.

Decision-Based Branching

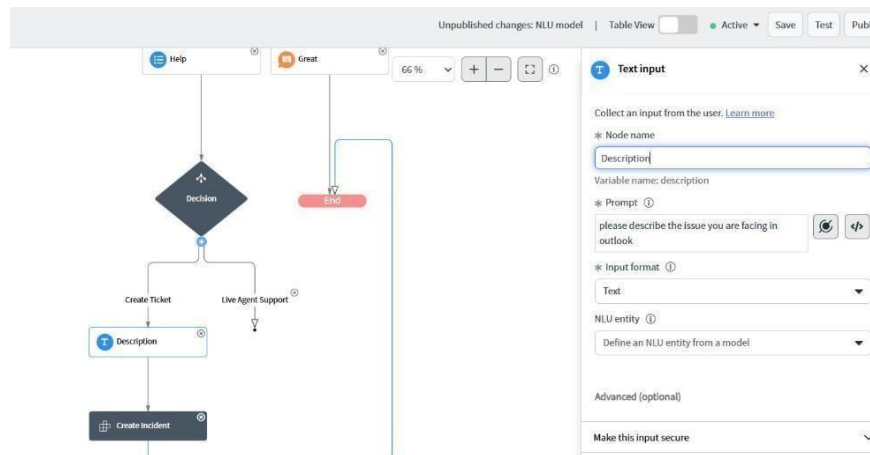
- If the user selects **"Yes"**, the flow proceeds to the **"Great"** response, confirming resolution.
- If the user selects **"No"**, the flow redirects to **"Help"**, offering additional assistance.

Customization Options

- The right panel allows modifications, such as setting a **default value**, enabling **confirmation messages**, and handling **conversation switching**.



- Or, we can also raised a ticket about this issue.



- If the issue remains unresolved, a ticket can be raised for further assistance. When the Virtual Agent is unable to solve the problem, it automatically connects the user to a Live Virtual Agent for real-time support through the Manage Live Chat Support feature.
- To handle this scenario, a new flow is created, allowing users to choose between "Create a Ticket" or "Live Agent Support." The flow is then configured using a Decision node to determine the appropriate action. Finally, a Topic Block is added from the Designer to integrate the process seamlessly.

5.3 Creating Topic Block

Topic Block: In ServiceNow Virtual Agent, a topic block is a reusable topic that can be called or invoked by other topics, allowing for modular and efficient chatbot development.

Creating a Topic Block for "Create a Ticket"

To streamline the ticket creation process, we are setting up a **Topic Block** specifically for handling user requests to raise a ticket. There are two ways to implement this:

1. Using an Existing Node:

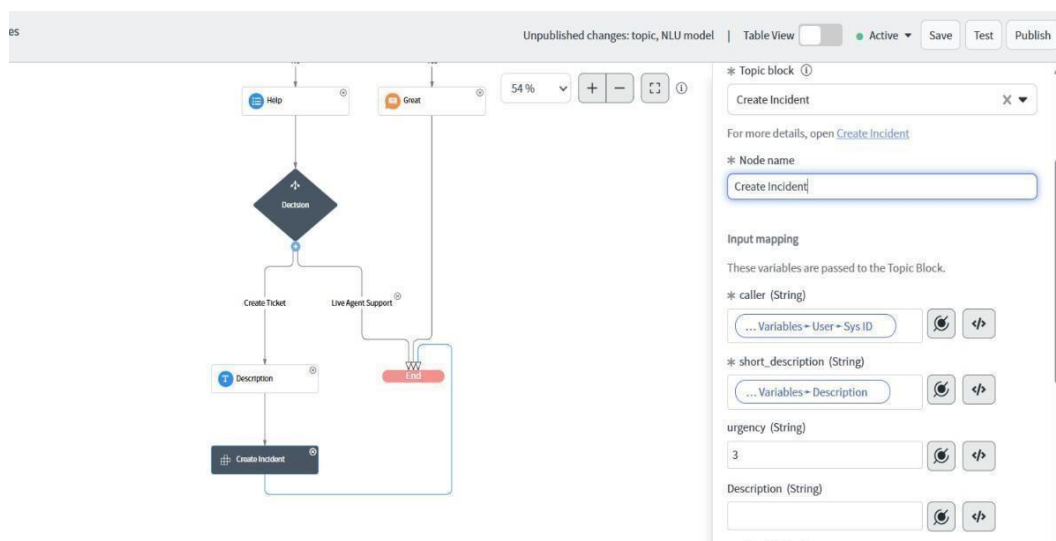
- If a predefined node already handles ticket creation, we can simply integrate it into the topic block to maintain consistency and reduce development effort.

2. Creating a New Node:

- If no existing node fits the requirement, a new node can be created within the topic block, customized to gather all necessary details from the user.

How the Topic Block Works

- When a user chooses to "**Create a Ticket**", the Virtual Agent initiates the topic block.
- The chatbot then prompts the user for a **description** of the issue.
- The **Topic Block** captures the description and any additional details required.
- The information is then processed to automatically generate a **ticket** in ServiceNow based on the user's input.
- When the user wants to create a ticket the virtual agent asks for the description, The topic block takes the description from the user and raises a ticket with the user requirements.
- Complete the flow and connect endings.



Positive Acknowledgment ("Great" Node)

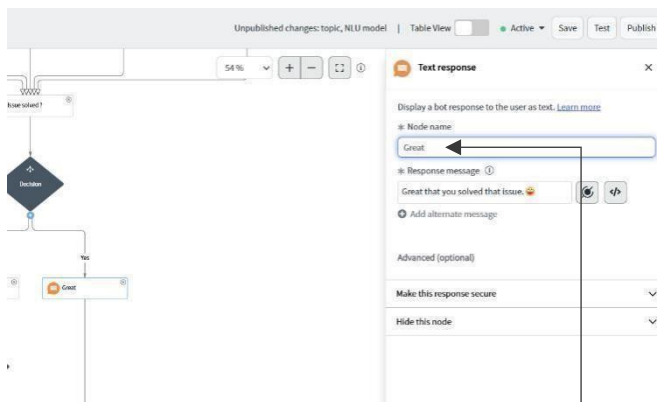
If the user selects "**Yes**", the bot responds with a **text message** saying,
“Great that you solved that issue. 🎉”

This message provides a **friendly and positive** user experience, reinforcing

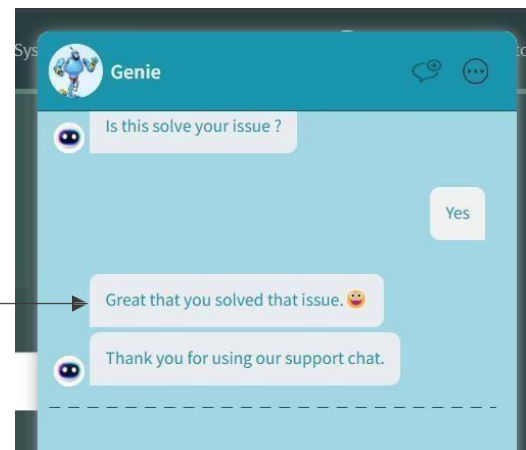
successful resolution.

Customization Options

- The right panel allows adding **alternate messages** for variation in responses.
- There are **advanced options** to secure or hide the response node as needed.



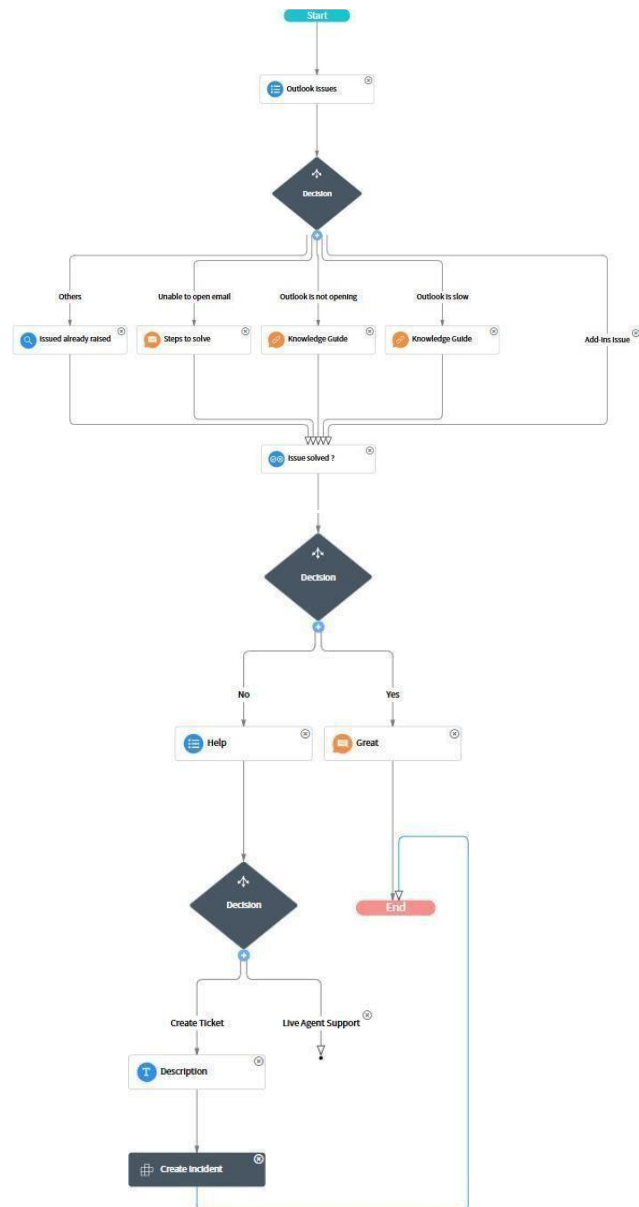
Bot Response



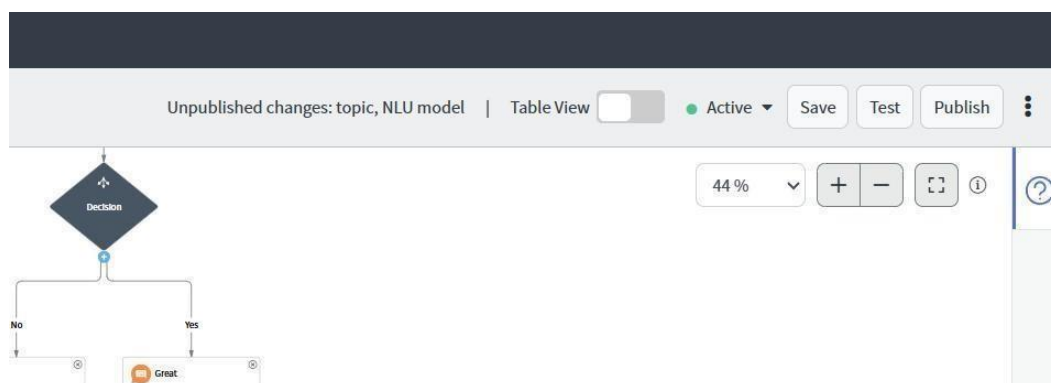
The Outlook Issue Resolution Flow in ServiceNow Virtual Agent starts when a user reports an issue. The chatbot categorizes it using a Decision Node with options like Others, Unable to Open Email, Outlook Not Opening, Outlook is Slow, and Add-ins Issue.

If the issue is already raised, the bot checks for previous reports; otherwise, it provides step-by-step guidance or directs users to a Knowledge Guide for self-service. Once troubleshooting is completed, the bot asks, "Issue solved?" If resolved, the conversation ends; if not, further support is provided.

5.4 Final Flow Chart:



- Click on “**Save**”



- Click on “**Publish**”.

CHAPTER 6

TOPICS

6.1 Topic : Genie



Start Node

- The flow begins from the "Start" node, initiating the chatbot process.

Welcome Message (Text Response Node)

- The chatbot sends a welcome message:
 - Hello 😊
 - This is Genie.
- This sets a friendly tone for user engagement.

User Input (Text Node - "prompt")

- The bot collects input from the user, likely to determine the request.

Processing

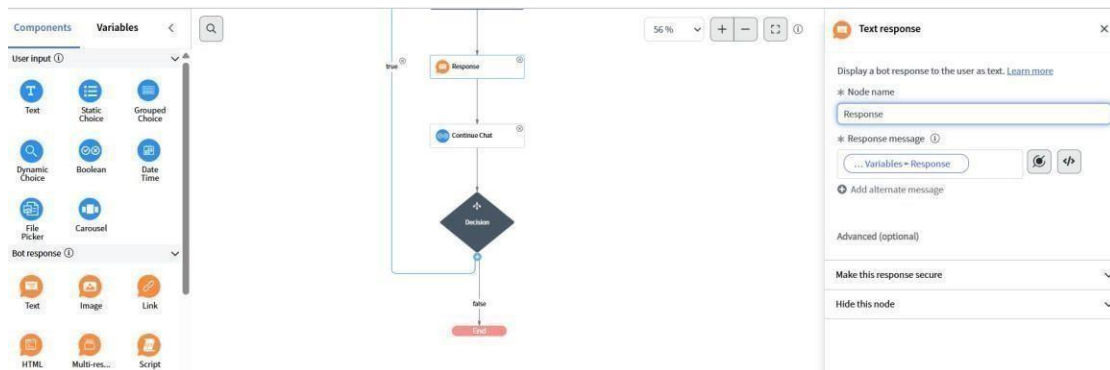
- The chatbot uses a "Gemini Script" to process the user's prompt.

Response Node (Text Response)

- After processing the user input, the chatbot sends a response.

Output Screen





Response Node:

- This node delivers a response to the user, which is dynamically fetched from a variable (Variables → Response).
- This means the response is likely generated based on user input or an AI script.

(Dynamic Content - Based on User Query)

- If user asks about weather:
"The current weather in your area is sunny with a temperature of 25°C."
- If user asks for support:
"I see you're facing an issue. Would you like to talk to our support team? (Yes/No)"
- If user asks for general help:
"Sure! Here are some common topics I can assist with: Billing, Technical Issues, General Questions."

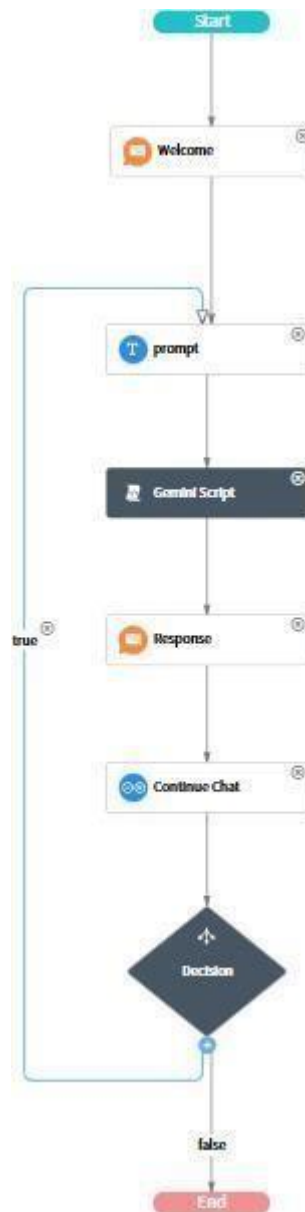
Continue Chat Node:

- After delivering the response, the chatbot has an option to continue the conversation.

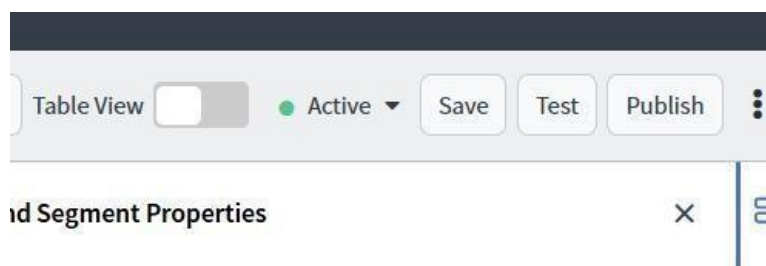
Decision Node:

- The flow contains a decision point that determines whether to continue the chat (true) or end the session (false).
- If false, the conversation ends.

6.1.1 Final Flow Chart



Click on “save” then click on “Publish” Button.



CHAPTER 7

NATURAL LANGUAGE UNDERSTANDING (NLU)

Natural Language Understanding (NLU) plays a crucial role in enhancing the conversational capabilities of virtual agents. It allows them to interpret user inputs, understand intent, and extract relevant information, making interactions more seamless and intuitive.

7.1 What is an NLU Model ?

An NLU model provides critical insights that help a virtual agent determine:

- What users want to accomplish (Intent Recognition)
- The specific details they provide (Entity Extraction)
- The different ways users phrase their requests (Utterance Variability)

By leveraging NLU, virtual agents can process human language more effectively, resulting in a more engaging and natural conversational experience.

7.2 Key Components of an NLU Model

An NLU model typically consists of the following three core elements:

1. Intents

An intent represents the goal or purpose of a user's message.

It defines what action the user wants to perform.

For example, a virtual agent for IT support may have the following intents:

- "Submit a service ticket" – when a user wants to report a technical issue.
- "Check order status" – when a user wants to track an order.
- "Request password reset" – when a user is having trouble logging in.

Example Intent:

- User Input: "Outlook not responding."
- Identified Intent: "Report an email issue"

2. Utterances

Utterances are the different ways users might express the same intent. Since users phrase their requests in various ways, NLU models must recognize multiple possible expressions of an intent.

Examples of Utterances for the Intent "Report an email issue":

- "Having issues with my Outlook."
- "Unable to send an email."
- "Outlook is not working properly."
- "Email service is down."

By training an NLU model with a diverse set of utterances, virtual agents become more adaptable to different ways users phrase their requests.

3. Entities

Entities provide additional details related to a user's request. They act as contextual parameters that help the virtual agent understand what the user is referring to.

Types of Entities:

- Objects: Specific items or services (e.g., "laptop," "printer," "email account").
- Identifiers: Unique values like case numbers, order numbers, or user IDs.
- Locations: Places related to the request (e.g., "New York office").
- Dates & Times: Time-related information (e.g., "tomorrow at 3 PM"). Example

with Entity Extraction:

- User Input: "I need a service ticket for my broken laptop."
- Intent: "Submit a service ticket"
- Entity Identified: Laptop (Device type)

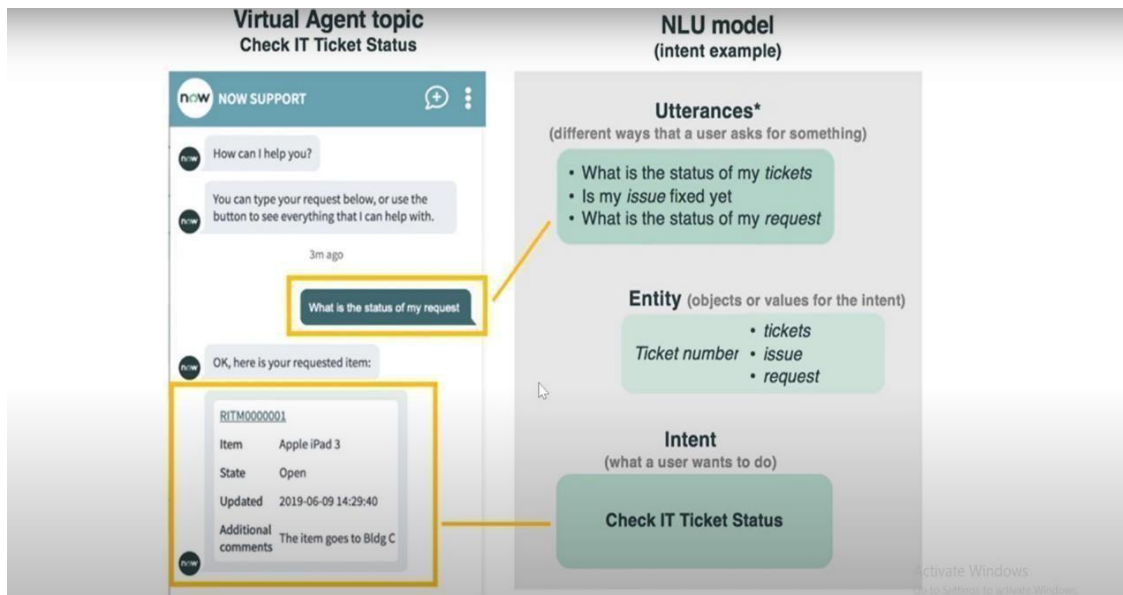


Fig 7.1 Nlu

Understanding NLU in Virtual Agents: IT Ticket Status Use Case

The image demonstrates how Natural Language Understanding (NLU) enables virtual agents to process user queries related to IT ticket statuses. It highlights how an NLU model identifies user intent, extracts relevant entities, and matches utterances to provide accurate responses.

Breakdown of the NLU Model for IT Ticket Status

1. User Query (Utterance Example)

- "What is the status of my request?"
- The virtual agent processes this input to understand the user's intent.

2. Intent Recognition: "Check IT Ticket Status"

- The system identifies that the user wants to retrieve the current status of an IT ticket.

3. Entity Extraction

- The model extracts key details, such as:
 - Ticket Number (e.g., RITM0000001)
 - Issue Type (e.g., iPad repair)

- Request Status (e.g., Open, In Progress, Resolved)

4. Response Generation

- The virtual agent fetches ticket details and provides an informative response:
 - Item: Apple iPad 3
 - State: Open
 - Updated: 2019-06-09 14:29:40

7.3 Creating an NLU Model

To set up a Natural Language Understanding (NLU) model, follow these steps:

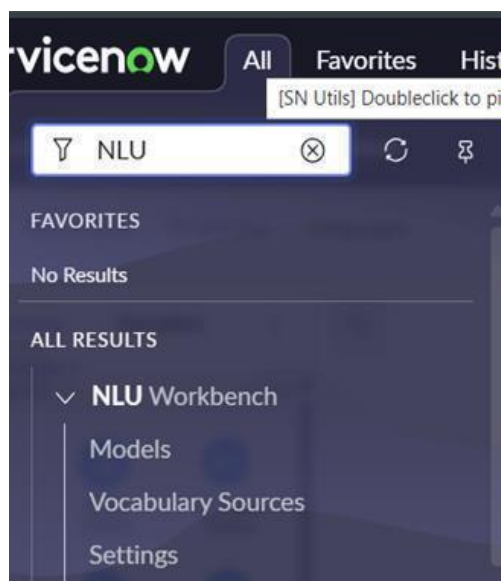
1. Navigate to the NLU Models Section

Go to **Application Navigator** > **NLU Models**

Click on "**Models**" to access the NLU model creation interface.

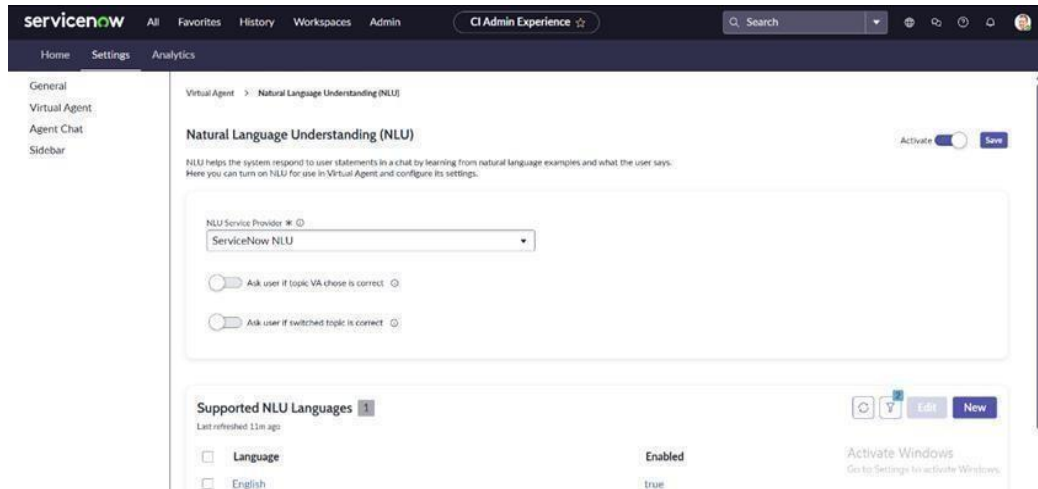
2. Set Up NLU in the Conversational Interface

Configure the NLU model within the conversational AI platform to enable natural language proc



3. Activate the NLU Model

Once the setup is complete, activate the model to start processing user queries and extracting meaningful insights.



Creating an NLU Model from Topic Properties

1. Selecting or Creating an NLU Model

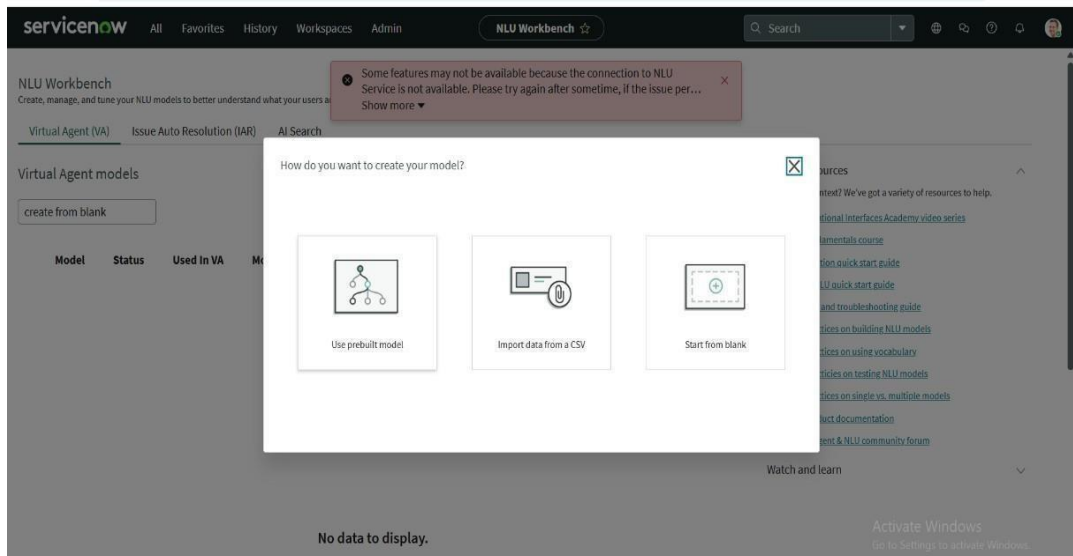
- From the Topic Properties, choose an existing NLU model.
- Alternatively, create a new model tailored to your specific requirements.

2. Creating a New NLU Model

- If a suitable model is not available, you can develop a custom NLU model to better fit your use case.

3. Accessing the NLU Workbench

- Navigate to All in the application menu.
- Search for "NLU Workbench" to access tools for building and managing your NLU models.

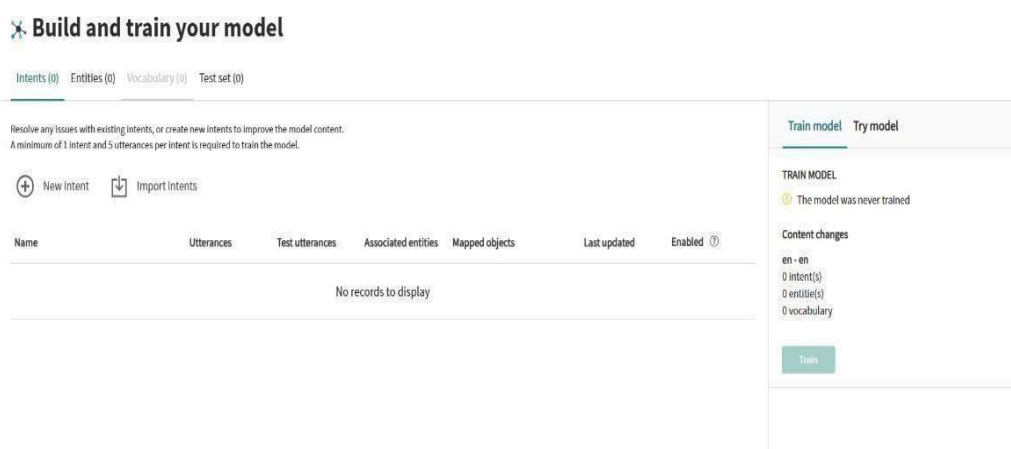


4. Create a Blank Model

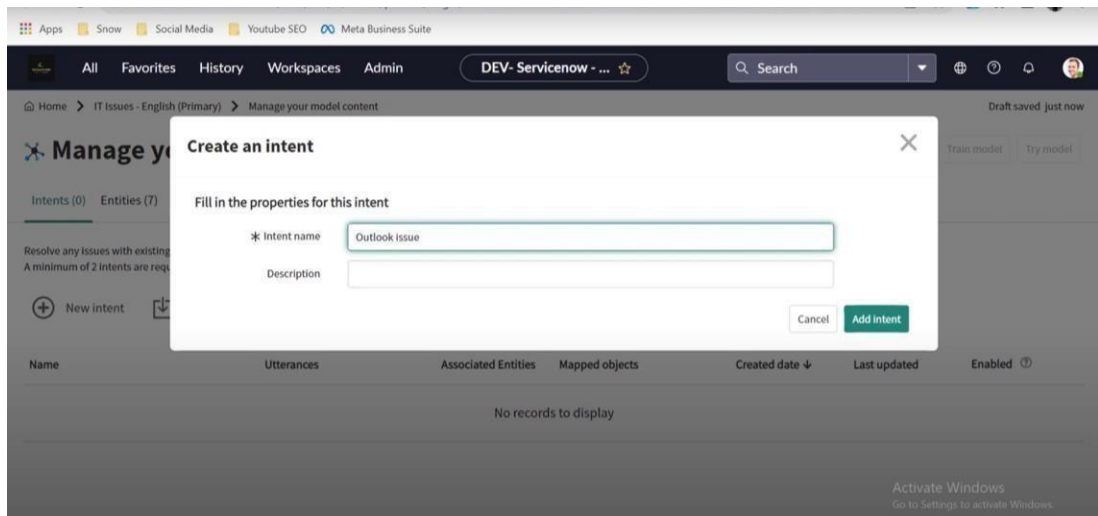
- Click on "Start from Blank" to initiate a new NLU model.
- Fill in the required details, such as the model name and description.

5. Add Content to the Model

- Click on "View Model" to open the model interface.
- Since the model is empty, add necessary content, such as intents, utterances, and entities, to train it effectively.



Click on “**New Intent**”.



Steps to Create Utterances for an Intent

1. Navigate to the Intent Section

- Once you have created and named your intent (e.g., "Outlook issue"), you will be directed to the intent details page.

2. Add Utterances

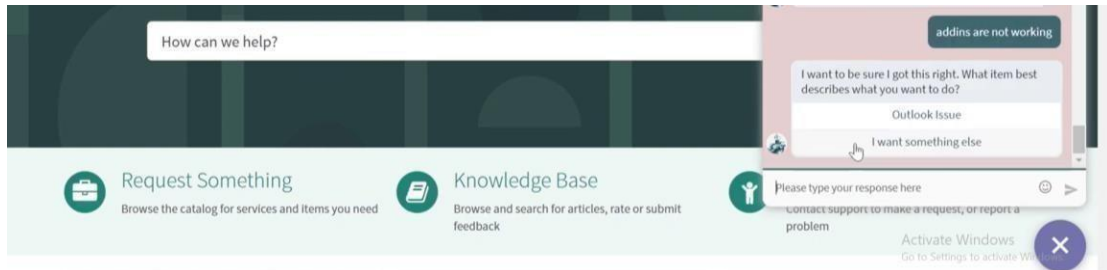
- Click on the "New Utterance" button.
- Type the utterance, which represents how users might phrase their issues. Examples for an "Outlook issue" intent could include:
 - "My Outlook is not opening."
 - "I am unable to send emails in Outlook."
 - "Outlook keeps crashing."
 - "Addins is not working."



3. Save the Utterances

- Click on the "Add" button after typing each utterance. Repeat the process to create multiple utterances.

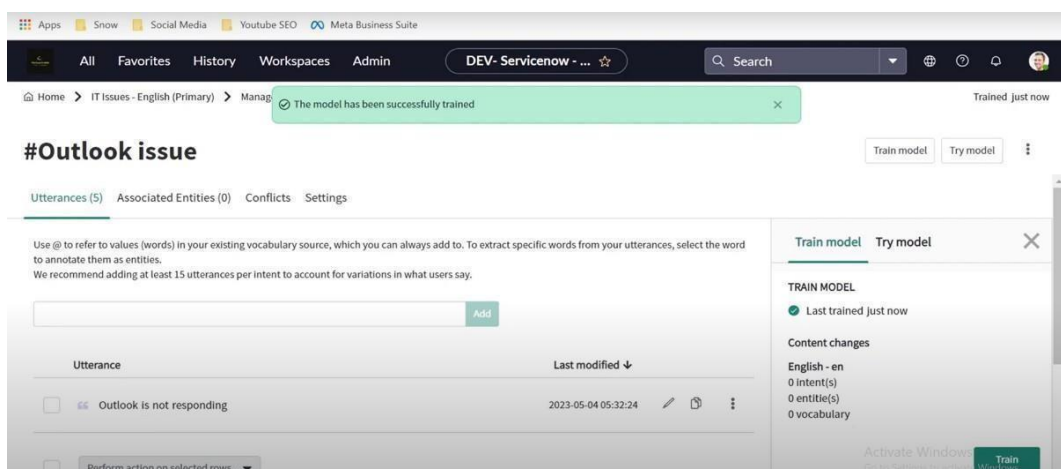
Creating these utterances allows the virtual agent to match user-provided keywords with the relevant issue automatically.



Improving Model Accuracy

- Training the model with various utterances ensures better recognition of user queries.
- The more diverse the utterances, the more accurately the agent can respond to different phrasings of the same issue.

To train our NLU models atleast 2 intents need to be created. The another intent is also related to the IT issues.



4. Test the Model

- After training, use the "Try Model" option to check if your intent

is correctly recognized based on the utterances.

5. Deploy the Model

- If the results are satisfactory, proceed to deploy the model for use in the chatbot or virtual agent.

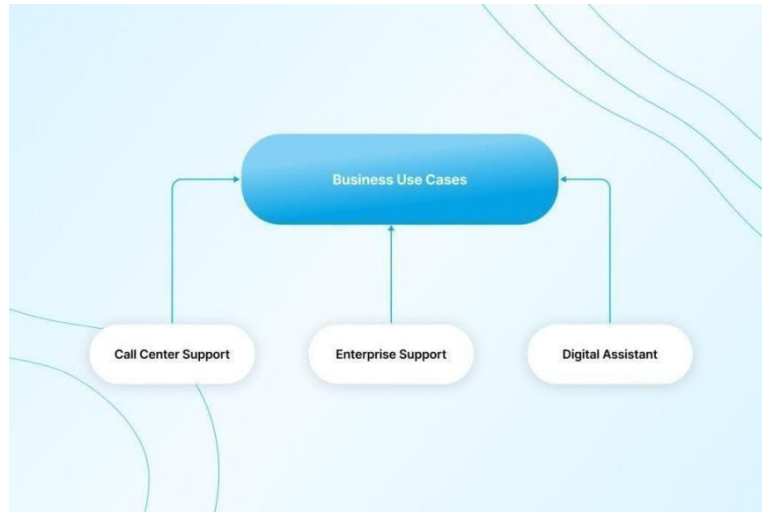
Setting up a Natural Language Understanding (NLU) model is a critical step in enhancing the capabilities of a conversational interface. By following a structured approach—from accessing the NLU Models section and configuring your model, to training it with well-defined intents and diverse utterances—you can build a model that accurately understands and processes user queries. Creating multiple intents and populating them with realistic utterances ensures the model is robust and reliable.

Utilizing tools like the NLU Workbench makes it easier to manage and refine your model, while the testing and deployment phases confirm its readiness for real-world interaction. Ultimately, a well-trained NLU model enables your virtual agent to deliver accurate, context-aware responses, significantly improving user experience and support efficiency.

CHAPTER 8

USE CASE DIAGRAMS

8.1 Business Use Case Diagram



Call Center Support

If your company has a call center that drains a lot of money from the budget, a ServiceNow chatbot can be quite useful. You can significantly reduce operational costs and enhance the user experience by building even a basic AI-powered chatbot. Especially when dealing with mundane tasks like scheduling, password resetting, and providing basic company information, chatbots will help live agents automate these requests and focus on more challenging topics.

Enterprise Support

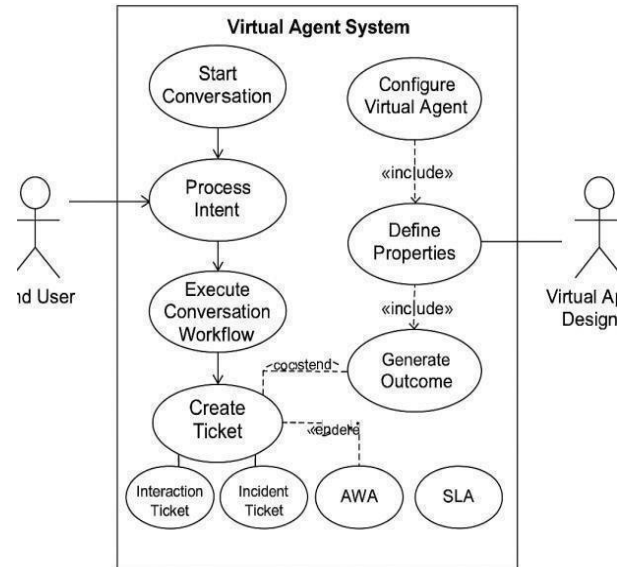
Similar to customer support, employee support can also benefit from request automation. Instead of spending time looking for a tool or information, employees can simply chat with a bot and immediately receive helpful links. Virtual agents can also streamline incidents and speed up the decision-making process for employees.

Digital Assistant

A more advanced ServiceNow chatbot can go beyond basic inquiries and perform sophisticated tasks such as event booking or personalized recommendations. Using predictive intelligence and [data analytics](#), virtual assistants can anticipate customer needs and understand the intent behind natural language. These types of chatbots are

also more inclusive and help users with disabilities navigate your website without any additional assistance.

8.2 Virtual Agent System Use Case Diagram



The Virtual Agent System enables end users to interact through a messaging interface to get support, trigger workflows, or create tickets. The system intelligently processes input using Natural Language Understanding (NLU) to determine intent and route conversations accordingly.

Actors

- End User: Initiates and interacts with the Virtual Agent.
- Virtual Agent Designer: Develops and configures conversation flows, NLU models, and properties.

Key Use Cases

- Start Conversation: User opens a chat via the messaging interface.
- Process Intent: The Virtual Agent uses NLU to recognize user intent.
- Execute Conversation Workflow: Based on intent, a predefined process is triggered.
- Generate Outcome: The system provides a response or action (like creating a ticket).

- Create Ticket: Depending on the case, the system can raise:
 - Interaction Ticket, Incident Ticket, AWA , SLA
- Configure Virtual Agent: The designer sets up conversation topics, properties, and logic.

Relationships

- Ticket creation may extend into various ticket types

The integration of ServiceNow's Virtual Agent into various business scenarios offers significant value by streamlining operations, reducing costs, and enhancing user satisfaction. From call center support and enterprise support to functioning as an intelligent digital assistant, virtual agents enable automation of routine tasks and deliver instant, personalized responses. The Virtual Agent System Use Case Diagram further illustrates how users, designers, and backend workflows interact seamlessly using Natural Language Understanding (NLU) to interpret intent and trigger appropriate actions. By leveraging these capabilities, organizations can foster more efficient service delivery, empower both employees and customers, and ensure scalable support infrastructure for future growth.

CHAPTER 9

TESTING AND DEPLOYMENT

9.1 Testing the Virtual Agent

Once the Natural Language Understanding (NLU) model has been successfully created, along with the defined intents and sample utterances, the next step is to thoroughly test the virtual agent to ensure it can accurately interpret user input and respond appropriately.

Test Scenario

User Input: “My Outlook stopped receiving emails.”

Expected Behaviour of the Virtual Agent:

1. Intent Recognition:

The virtual agent should correctly identify the user’s issue as related to the “Outlook Email Issues” intent (or similar, based on how the intent is named in the model).

2. Response Generation:

Based on the recognized intent, the virtual agent should respond with a relevant troubleshooting message or offer helpful steps. For example:

Virtual Agent Response:

"I'm sorry to hear you're having trouble with Outlook. Let's try a few steps to resolve this issue:

1. Please check your internet connection.
2. Ensure Outlook is connected — look for the 'Connected to Server' status.
3. Try restarting Outlook and your device.
4. Check if your mailbox is full or if there are any sync errors.

Would you like me to guide you through any of these steps?"

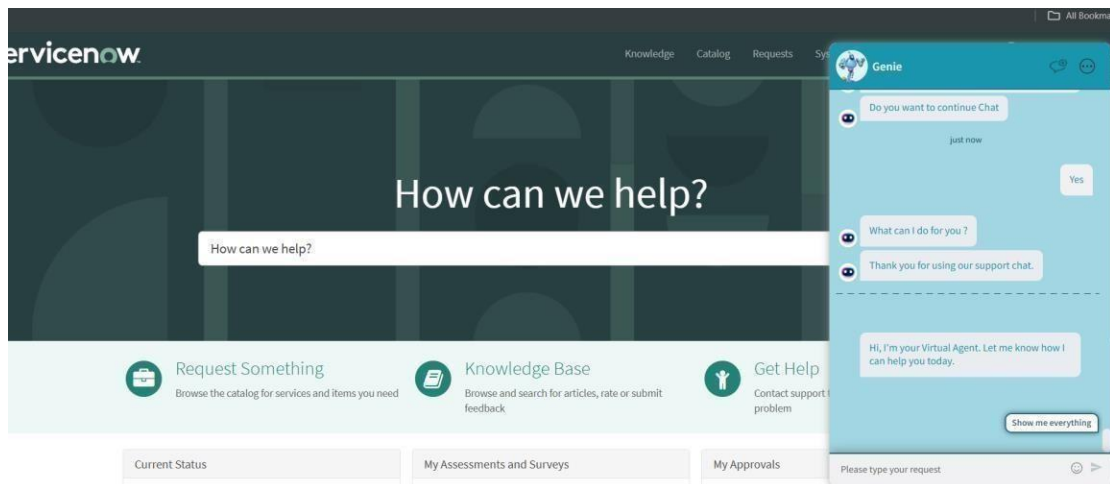
3. Fallback Handling (if applicable):

If the virtual agent fails to identify the intent correctly, it should trigger a fallback response such as:

"I'm not sure I understood that completely. Are you having trouble receiving emails in Outlook?"

Purpose of Testing:

- To verify that the virtual agent correctly understands and classifies user queries.
- To ensure the responses are helpful, relevant, and guide the user effectively.



- Then the users finds the efficient way to resolve the issue.



9.2 Deployment of the Virtual Agent

Once development and testing are complete, the virtual agent is ready for deployment to the production environment.

Pre-Deployment Checklist

Ensure the NLU model is trained, all intents and responses are finalized, testing is complete, and system integrations are functioning. Set up required credentials and security configurations.

Deployment Process

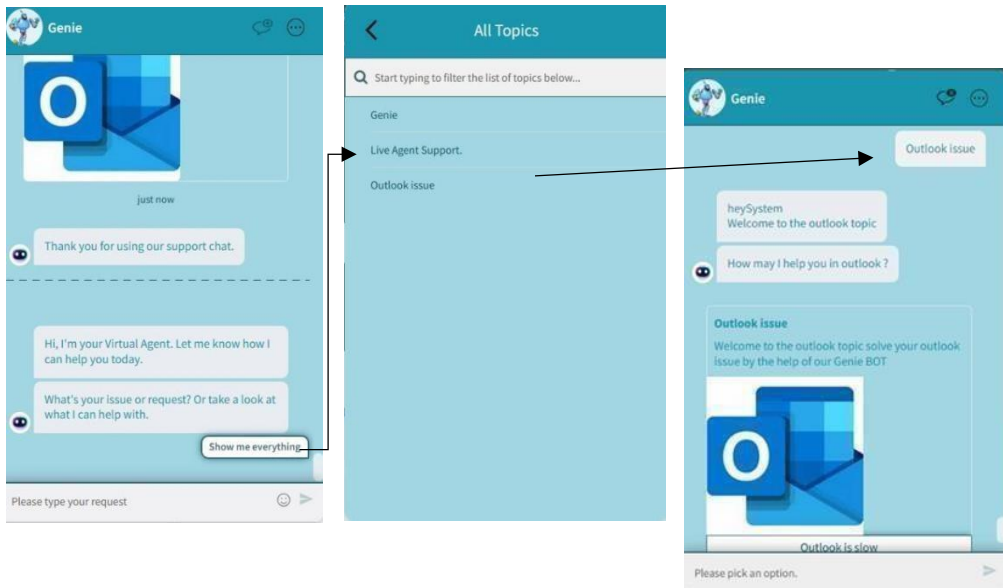
1. **Publish the Bot:** Deploy the model and conversation flows using the virtual agent platform (e.g., Power Virtual Agents, Dialogflow).
2. **Connect to Channels:** Integrate the bot with platforms like Microsoft Teams, web chat, or mobile apps.
3. **Enable Monitoring:** Set up logs, usage analytics, and fallback/error tracking to monitor performance.
4. **Access Control:** Define user access and perform a soft launch if needed.

9.3 Output Screens

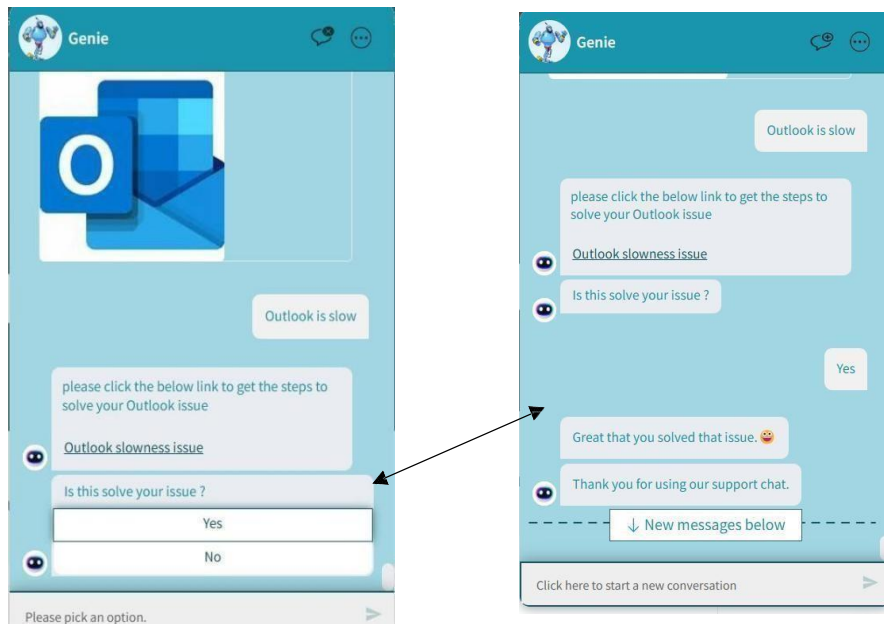
The output screen is the user-facing interface where the virtual agent interacts with users in real time. It displays the conversation flow, including the user's queries and the virtual agent's responses.

Key Elements of the Output Screen:

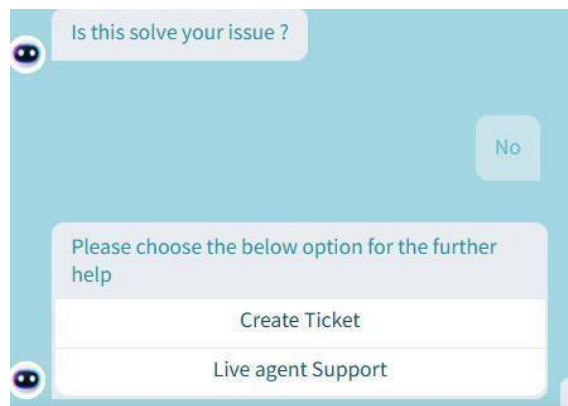
- **Chat Interface:**
A visually engaging chat window where users can type queries and view responses from the virtual agent.
- **User Input Area:**
A text box (or voice input option) where users enter their messages.
- **Virtual Agent Responses:**
Displays dynamic replies based on recognized intents. Responses may include text, buttons (e.g., “Yes / No”), quick replies, or links.
- **Action Triggers (if applicable):**
The screen may also show triggered actions like creating a ticket, sending an email, or retrieving account details.



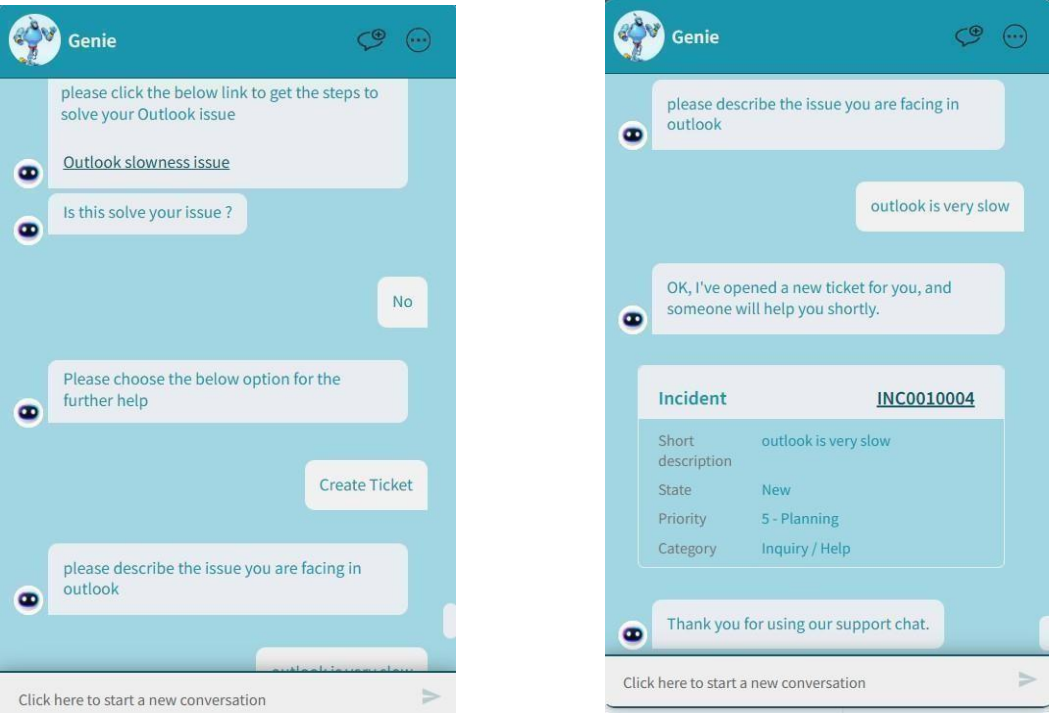
When we click on “**Outlook is slow**” the output as



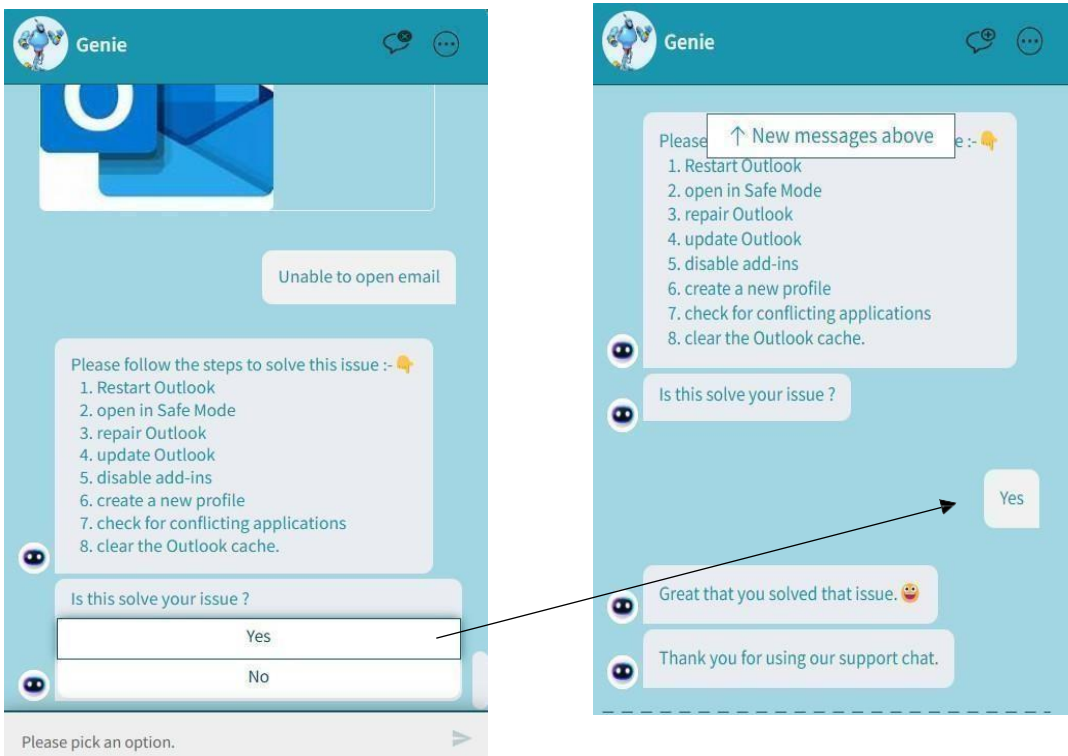
If “**No**”.



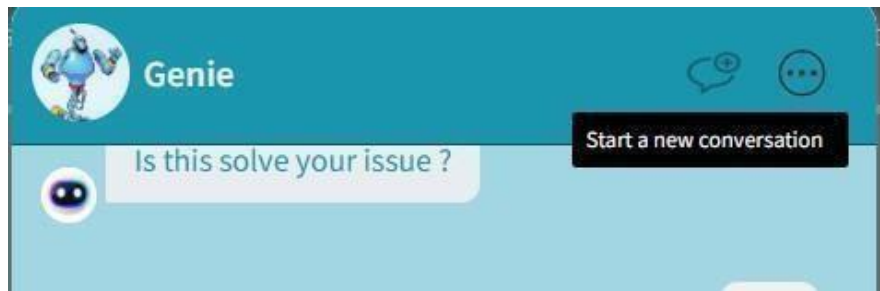
It will generate a Incident Record Automatically with a INC_____.



When we click on “Unable to open email” the output as



Start “**New Conversation**”.



Chat Window:

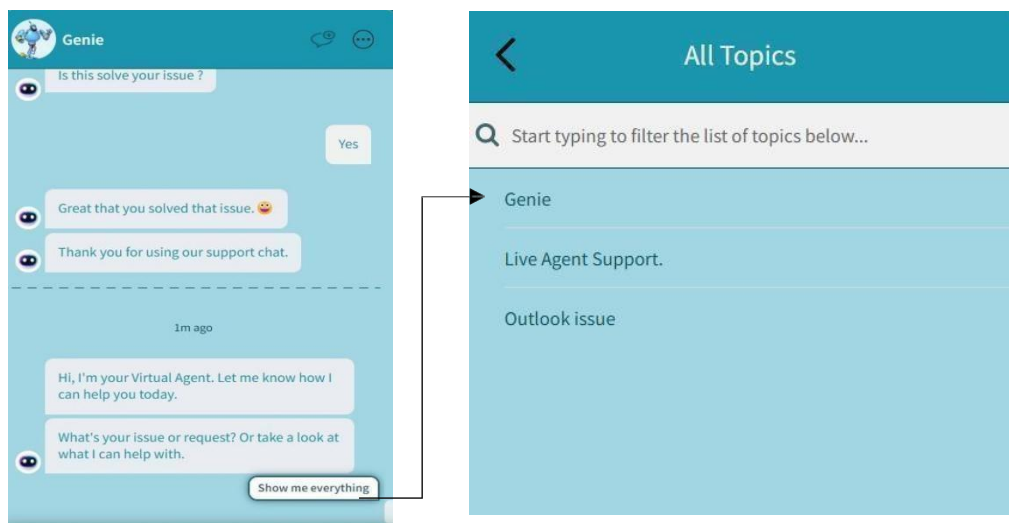
- The conversation is structured in a chat format.
- The virtual agent (Genie) displays messages in light blue chat bubbles.
- User responses are shown separately for easy readability.

Interactive Responses:

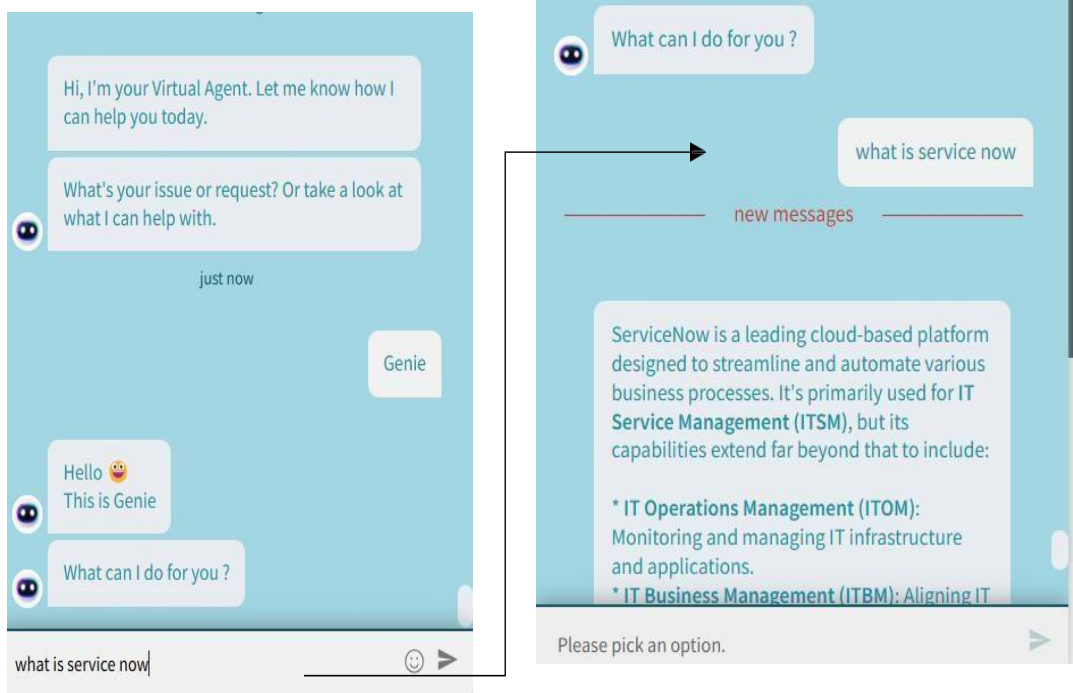
- The virtual agent acknowledges user input.
- It provides a positive and engaging response (e.g., using emojis to enhance interaction).

Quick Action Buttons:

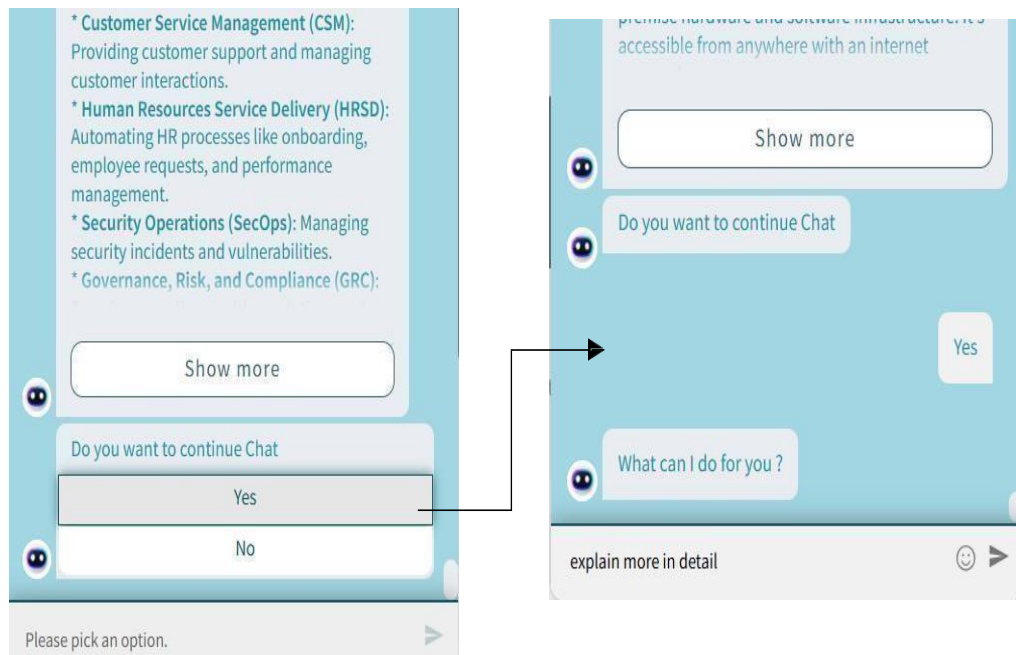
- A button labeled “**Show me everything**” allows users to explore more option

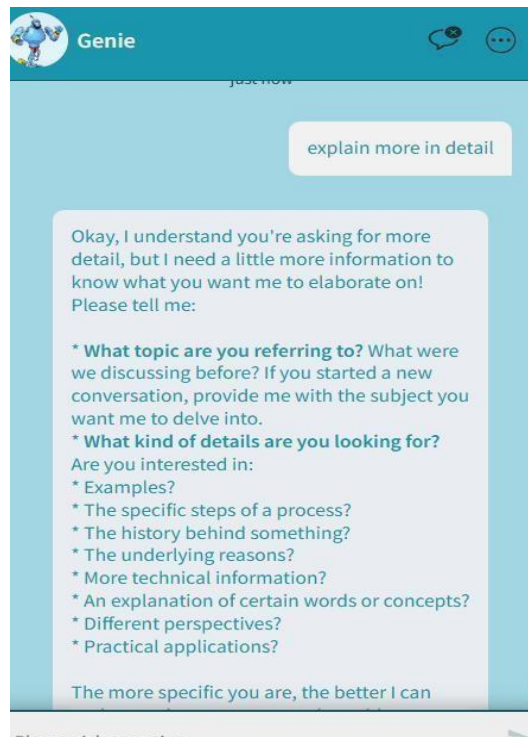


By clicking on “Genie”.

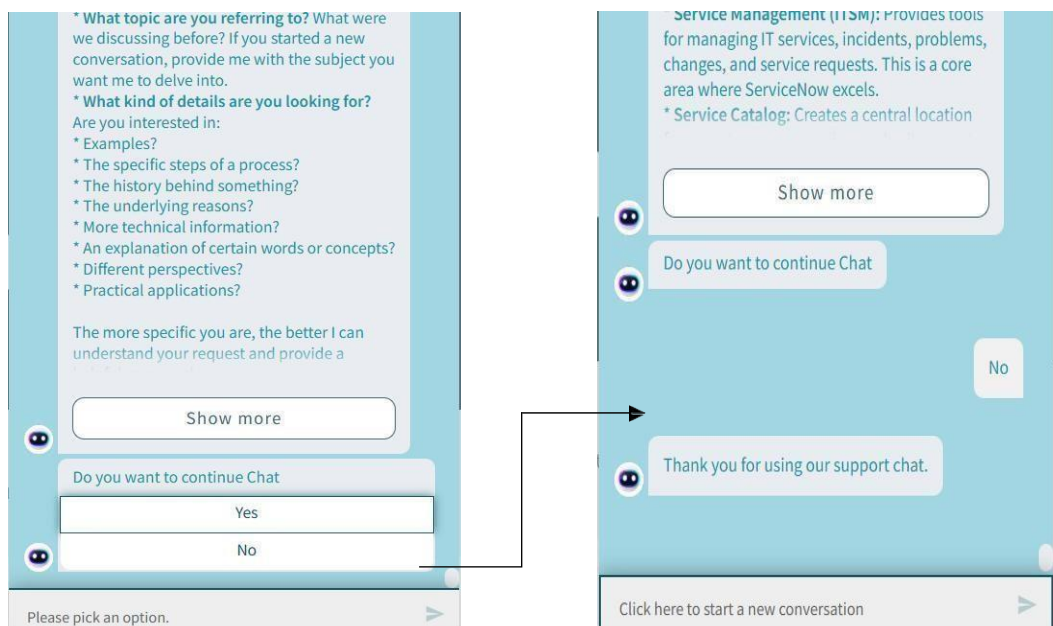


By clicking on “Yes”.

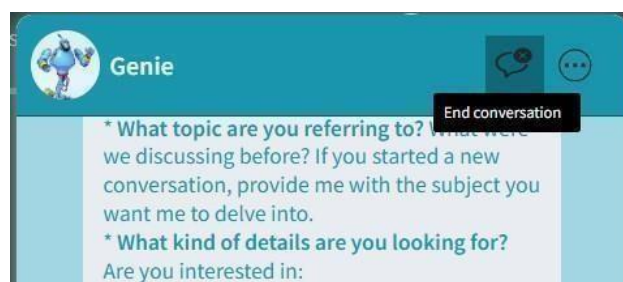




Clicking on “NO”.



To End a conversation



9.4 Difference between ChatGPT and Service now Virtual Agent :

ServiceNow Virtual Agent and ChatGPT are both AI-driven chat solutions but serve different purposes. ServiceNow Virtual Agent is designed for ITSM, HR, and customer support automation within the ServiceNow platform. It uses intent-based NLU to guide users through structured workflows, such as troubleshooting IT issues, handling HR inquiries, and automating ticket creation. It integrates directly with ServiceNow modules like ITSM and the Knowledge Base, ensuring accurate and process-driven responses.

In contrast, ChatGPT is a general-purpose AI chatbot specializing in free-form, generative conversations. It assists with content creation, brainstorming, coding, and research, generating responses dynamically instead of following predefined workflows. While ChatGPT can integrate with various platforms via APIs, it does not natively connect to enterprise systems like ServiceNow.

Feature	ChatGPT	ServiceNow Virtual Agent
Purpose	General-purpose AI for answering queries, conversations, and content generation.	Enterprise-focused chatbot for automating IT, HR, and business service requests.
Technology	Uses OpenAI's GPT model for natural language processing (NLP) and contextual understanding.	Uses AI, NLU, and rule-based workflows specific to ServiceNow's ITSM, HR, and customer service functions
Customization	Pre-trained with vast general knowledge but can be fine-tuned with API-based integrations	Highly customizable within ServiceNow, allowing custom topics, flows, and enterprise integrations
Integration	Works as a standalone AI but can integrate with third-party apps via APIs.	Seamlessly integrates with ServiceNow ITSM, HR, CSM, and enterprise workflows.

Automation Capabilities	Primarily assists with text-based responses and content creation.	Automates incident management, service requests, ticketing, troubleshooting, and more.
User Interaction	Designed for open-ended, conversational AI experiences.	Focused on structured enterprise support with guided responses and predefined workflows.
Scalability & Use Case	Best for customer engagement, creative tasks, and general knowledge queries.	Best for business process automation , reducing IT/HR workloads, and handling enterprise queries.

CHAPTER 10

CONCLUSION

10.1 Conclusion

The implementation of the ServiceNow Virtual Agent significantly enhances IT and business support operations by automating routine tasks, resolving common user issues, and streamlining service delivery. Through this project, we explored the architecture and capabilities of ServiceNow's AI-driven chatbot, its integration with enterprise systems, and its impact on organizational efficiency.

The results demonstrate that virtual agents can reduce response times, improve user satisfaction, and free up valuable human resources to focus on more complex tasks. Our findings suggest that with continued development and customization, ServiceNow Virtual Agent can play a vital role in the digital transformation journey of any modern enterprise.

This project has not only deepened our understanding of AI-powered service management but also equipped us with practical experience in leveraging automation for scalable, intelligent business solutions.

- **Enhanced Efficiency** – Automates routine tasks, reducing workload and response time.
- **Improved User Satisfaction** – Provides **24/7 support** with quick, intelligent responses.
- **Optimized Business Operations** – Seamless integration with ServiceNow's IT, HR.
- **Cost Savings** – Reduces operational expenses by minimizing human intervention.
- **Scalability and Flexibility** – Easily adaptable to evolving business needs and workflows.

10.2 Summary of Benefits and Improvements

1. **24/7 Automated Support** : The Virtual Agent provides around-the-clock assistance, ensuring users get help at any time without human intervention.
2. **Reduced Workload on Support Teams** : By handling repetitive and low-complexity queries, it frees up IT and HR teams to focus on more critical and high-priority tasks.
3. **Faster Response and Resolution Times** : Automated responses to common issues like password resets or ticket status checks drastically reduce the time users spend waiting for help.
4. **Consistency in Service Delivery** : Virtual Agents deliver consistent answers and processes, reducing variability and human error in responses.
5. **Improved User Satisfaction** : Users benefit from immediate help, which enhances their experience and satisfaction with IT and business services.
6. **Scalability** : As organizations grow, Virtual Agents can handle increasing query volumes without the need for proportional increases in support staff.
7. **Seamless Integration with ServiceNow Platform** : The Virtual Agent integrates easily with existing ServiceNow modules, leveraging workflows, knowledge bases, and incident management tools.
8. **Cost Efficiency** : Automating common tasks and reducing ticket volume leads to lower operational costs over time.
9. **Data-Driven Insights** : Analytics from user interactions help organizations identify common pain points and improve services proactively.
10. **Enhanced Employee Productivity** : Employees spend less time waiting for support, enabling them to stay focused on their core responsibilities.

10.3 Future Enhancements

To further enhance its capabilities, the **Virtual Agent can be expanded** with:

1. **Advanced Natural Language Understanding (NLU)** : Improve the chatbot's ability to interpret complex queries using advanced NLU models, enabling it to handle more nuanced conversations.
2. **Multilingual Support** : Add support for multiple languages to serve a globally diverse user base and improve accessibility.
3. **Voice Integration** : Enable voice-based interactions by integrating with platforms like Amazon Alexa, Google Assistant, or speech-to-text APIs.
4. **Proactive Assistance** : Implement features that allow the Virtual Agent to offer help before users ask—e.g., by monitoring behavior and triggering prompts.
5. **Integration with More Business Applications** : Extend the Virtual Agent's functionality by connecting it to third-party tools such as Slack, Microsoft Teams, or enterprise CRMs and ERPs.
6. **Self-Learning Capabilities** : Incorporate machine learning to help the agent improve over time based on user interactions and feedback.
7. **Automated Workflow Creation** : Let the Virtual Agent suggest or even create new workflows based on repetitive tasks identified through usage analytics.
8. **Enhanced Security and Compliance Features** : Integrate robust authentication, data encryption, and audit logging to ensure compliance with organizational and legal standards.

Project Mapping with various courses of the Curriculum with attained POs

Name of Course from which Principles are applied in this project	Description of the Application	Attained POs
C311	Thorough Examination of Existing systems and definition of the problem	PO1,PO2, PO6,PO7
C324, C325, C415	Gathering, Analysis and classification of all the requirements for the proposed system	PO1,PO2,PO4
C425, C314, C225	Logical design is done by using logical tools	PO1, PO2,PO3, PO10
C228, C325, C324	The physical design is done by using Android Studio, and database MySQL.	PO5,PO8,PO10, PO11,PO12
C425, C324, C414	Each and every module is tested, integrated, and evaluated.	PO8,PO10,PO1 1,PO12
C425, C414	Implementation of the project in the real environment	PO7,PO8,PO10, PO11
C425, C3110, C329	Documentation is done by all the team members with well defined communication with inclusion of Engineering and management Principles	PO8,PO9,PO10, PO11,PO12
C425, C3110, C329	Presentation of the work in teams with proper method of presentation.	PO9,PO8,PO10, PO11

COURSE STRUCTURE (R13)

I YEAR- I SEMESTER

S No	Subject	T	P	Credits
C111	Problem Solving and Programming Using C	3	0	3
C112	Applied Chemistry	3	0	3
C113	Differential Equations	3	0	3
C114	Engineering Graphics	1	4	3
C115	Basics of Electrical and Electronics Engineering	3	0	3
C116	Problem Solving and Programming Using C Lab	0	3	1.5
C117	IT Workshop	0	3	1.5
C118	Applied Chemistry Lab	0	3	1.5
Total Credits				19.5

I YEAR- II SEMESTER

S No	Subject	T	P	Credits
C121	Communicative English	3	0	3
C122	Applied Physics	3	0	3
C123	Linear Algebra & Vector Calculus	3	0	3
C124	Digital Logic Design	3	0	3
C125	Python Programming	3	0	3
C126	Environmental Sciences	2	0	0
C127	Communicative English Skills Lab	0	3	1.5
C128	Applied Physics Lab	0	3	1.5
C129	Python Programming Lab	0	3	1.5
Total Credits				19.5

II YEAR- I SEMESTER

S No	Subject	T	P	Credits
C211	Probability & Statistics	3	0	3
C212	Mathematical Foundations of Computer Science	3	0	3
C213	Data Structures & Algorithms	3	0	3
C214	Object Oriented Programming through Java	3	0	3
C215	Introduction to Artificial Intelligence	3	0	3
C216	Constitution of India	2	0	0
C217	Data Structures & Algorithms Lab	0	3	1.5
C218	Object Oriented Programming through Java lab	0	3	1.5
C219	Introduction to Artificial Intelligence Lab	0	3	1.5
C2120	Skill Oriented Course -1	1	2	2.0
Total Credits				21.5

II YEAR- II SEMESTER

S No	Subject	T	P	Credits
C221	Numerical Methods & Transformations	3	0	0
C222	Computer Organization	3	0	0
C223	Database Management Systems	3	0	0
C224	Formal Languages and Automata Theory	3	0	0
C225	Managerial Economics and Financial Accountancy	3	0	0
C226	Database Management Systems Lab	0	0	3
C227	Web Application Development Lab	0	0	3
C228	R Programming Lab	0	0	3
C229	Skill Oriented Course -2	1	0	2
Total Credits				21.5

III YEAR- I SEMESTER

S No	Subject	T	P	Credits
C311	Design and Analysis of Algorithms	3	0	3
C312	Machine Learning	3	0	3
C313	Operating Systems	3	0	3
C314	Professional Elective -1	3	0	0
C315	Open Elective -1	3	0	3
C316	Skill Oriented Course – III	0	4	2
C317	Professional Ethics and Human Values	2	0	0
C318	Summer Internship one Month (Mandatory) after second year(to be evaluated during V Semester	0	0	1.5
C319	Machine Learning Lab	0	3	1.5
C3110	Operating Systems Lab	0	3	1.5
Total Credits				21.5

III YEAR- II SEMESTER

S No	Subject	T	P	Credits
C321	Computer Networks and communications	3	0	3
C322	Deep Learning	3	0	3
C323	Expert Systems	3	0	3
C324	Professional Elective -2	3	0	3
C325	Open Elective -2	3	3	3
C326	Skill Oriented Course – IV (Soft Skills)	0	2	2
C327	Intellectual property rights and patents (IPR&P)	2	0	0
C328	Deep Learning Lab	0	3	1.5
C329	Computer Networks and communications Lab	0	3	1.5
C3210	Mini Project with seminar	1	2	1.5
Total Credits				21.5

IV YEAR- I SEMESTER

S No	Subject	T	P	Credits
C411	Professional Elective-III	3	0	3
C412	Professional Elective-IV	3	0	3
C413	Professional Elective-V	3	0	3
C414	Open Elective-III	3	0	3
C415	Open Elective - IV	3	0	3
C416	Management Science	3	0	3
C417	Skill Oriented Course –V	0	4	2
C418	Industrial/Research Internship one months (Mandatory) after third year (to be evaluated during VII semester)	0	0	1.5
Total Credits				23

IV YEAR- II SEMESTER

S No	Subject	T	P	Credits
C421	Project work - Phase II	0	0	12
Total Credits				12

OPEN ELECTIVES		
Open Elective – 1 (V Semester) Python Programming	Open Elective -2 (VI Semester) Fundamentals of Artificial Intelligence	Open Elective -3 (VII Semester) Human Computer Interaction
Open Elective -4 (VII Semester) Applications of Artificial Intelligence	Skill Oriented Course (Advanced)– IV 1. (MEAN Stack Technologies - Module I- MongoDB, Express.js, Angular JS, Node.js and AJAX 2. Big Data : Apache Spark 3. DevOPS	
PROFESSIONAL ELECTIVES		
Professional Elective – 1 (V Semester)	Professional Elective – 2 (VI Semester)	Professional Elective – 3 (VII Semester)
1. Software Engineering 2. Compile Design 3. Data Visualization 4. Design and Analysis of Algorithms	1. Software Project Management 2. Distributed Systems 3. Internet of Things 4. Data Where Housing and Data Mining	1. Reinforcement Learning 2. Soft Computing 3. Cryptography and Network Security 4. NOSQL Databases 5. Natural Language Processing
Professional Elective – 4 (VII Semester)	Professional Elective – 5 (VII Semester)	
1. Robotic Process Automation 2. Cloud Computing 3. Big Data Analytics 4. Block Chain Technologies 5. Image & Video Analytics	1. Social Network Analysis 2. Recommender Systems 3. Computer vision 4. Object Oriented Analysis and Design 5. Semantic Web	

SANAVYA GONUGUNTLA

Successfully completed certification requirements for
Certified System Administrator

Issued: March 13, 2025 01:37:12 PM

Certification No. 26893074

Jayney Howson

Jayney Howson
Senior Vice President, Global Learning and Development
Servicenow

JONNALAGADDA NAGA SRI JYOTHI

Successfully completed certification requirements for
ServiceNow Certified System Administrator

Issued: March 10, 2025

Certification No. 26897712

Jayney Howson

Jayney Howson

Senior Vice President, Global Learning and Development
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